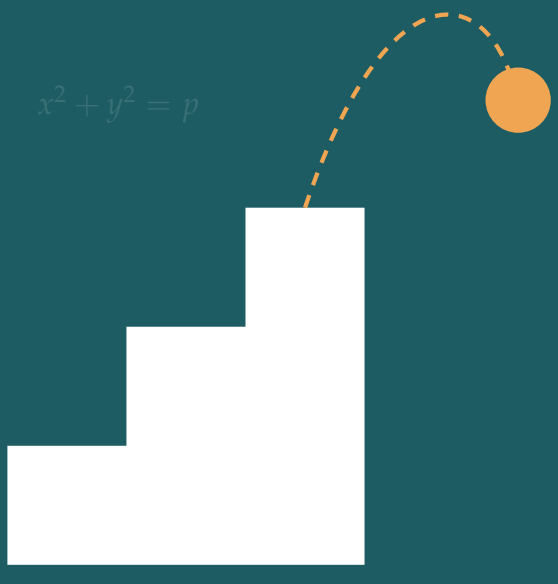
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP II AND III

2022 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP II AND III

2022 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2022 Worked Solutions

Worked solutions to all 24 questions of the 2022 STEP papers (STEP II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

Contents

STEP II

Q1 Integration by Parts Pure . . . 2
Q2 Recurrence Relations Pure . . . 6
Q3 Fibonacci Series and Decimals Pure . . . 9
Q4 Modulus Functions Pure . . . 13
Q5 Inequalities in a Triangle Pure . . . 17
Q6 Orthogonal Trajectories Pure . . . 21
Q7 Roots of Complex Polynomials Pure . . . 25
Q8 Invariant Lines of Matrices Pure . . . 28
Q9 Friction and Moments Mechanics . . . 32
Q10 Projectiles on Inclined Plane Mechanics . . . 36
Q11 Expected Loss Optimisation Probability and Statistics . . . 40
Q12 Mean, Median and Mode Probability and Statistics . . . 44

STEP III

Q1 Roots of Polynomials Pure . . . 48
Q2 Infinite Descent Pure . . . 52
Q3 Implicit Differentiation Pure . . . 56
Q4 Infinite Products Pure . . . 60
Q5 Symmetric Definite Integrals Pure . . . 64
Q6 Rolling Circles and Tangents Pure . . . 67
Q7 Vector Product Transformations Pure . . . 71
Q8 De Moivre and Irrationality Pure . . . 74
Q9 Collisions and Restitution Mechanics . . . 78
Q10 Elastic Springs Equilibrium Mechanics . . . 82
Q11 Binomial Mean Deviation Probability and Statistics . . . 85
Q12 Random Chords in a Circle Probability and Statistics . . . 89

WORKED SOLUTION

STEP II 2022, Question 1

Pure • Integration by Parts

STEP II 2022, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Every part rests on one move: integrating by parts produces a second integral that cancels, or nearly cancels, something you already have. In (i) and (ii) that means spotting that the integrand is the derivative of a product. In (iii) the same move shows that the difference between the two integrals is just the change in e^x/x between the limits, so the sketch in (a) does the rest of the work.

Solution

Part (i)

Integrate the first term by parts, with $u = \sqrt{1+x^3}$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$, so $v = x$ and $\frac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{3x^2}{2\sqrt{1+x^3}}$:

$$\begin{aligned}\int 2\sqrt{1+x^3}\,\mathrm{d}x &= 2x\sqrt{1+x^3} - 2\int x\cdot\frac{3x^2}{2\sqrt{1+x^3}}\,\mathrm{d}x \\ &= 2x\sqrt{1+x^3} - \int \frac{3x^3}{\sqrt{1+x^3}}\,\mathrm{d}x.\end{aligned}$$

The last integral is exactly the second term of the integrand, so adding it on cancels it:

$$\int\left(2\sqrt{1+x^3} + \frac{3x^3}{\sqrt{1+x^3}}\right)\mathrm{d}x = 2x\sqrt{1+x^3} + c.$$

Differentiating $2x\sqrt{1+x^3}$ by the product rule gives back $2\sqrt{1+x^3} + \dfrac{3x^3}{\sqrt{1+x^3}}$, which confirms it.

Part (ii)

Split the integrand as $\dfrac{\sin x}{x} + \dfrac{2\sin x}{x^3}$. Neither piece has an elementary antiderivative on its own (the first is the famous Si integrand), so they must combine, just as in (i). Integrate the second piece by parts, with $u = \sin x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = 2x^{-3}$, so $v = -x^{-2}$:

$$\int \frac{2\sin x}{x^3}\,\mathrm{d}x = -\frac{\sin x}{x^2} + \int \frac{\cos x}{x^2}\,\mathrm{d}x.$$

Now integrate $\dfrac{\cos x}{x^2}$ by parts, with $u = \cos x$ and $v = -x^{-1}$:

$$\int \frac{\cos x}{x^2}\,\mathrm{d}x = -\frac{\cos x}{x} - \int \frac{\sin x}{x}\,\mathrm{d}x.$$

The $\int \frac{\sin x}{x}\,\mathrm{d}x$ cancels against the first piece, leaving

$$\int (x^2+2)\frac{\sin x}{x^3}\,\mathrm{d}x = -\frac{\sin x}{x^2} - \frac{\cos x}{x} + c = -\frac{\sin x + x\cos x}{x^2} + c.$$

As a check, differentiating $-\dfrac{\sin x}{x^2}$ gives $-\dfrac{\cos x}{x^2} + \dfrac{2\sin x}{x^3}$, and differentiating $-\dfrac{\cos x}{x}$ gives $\dfrac{\sin x}{x} + \dfrac{\cos x}{x^2}$; the $\cos x/x^2$ terms cancel and the sum is the integrand.

Part (iii)(a)

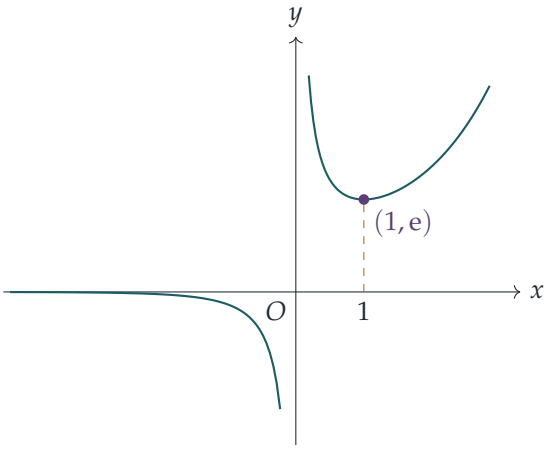
Let $y = \dfrac{\mathrm{e}^x}{x}$, defined for $x \neq 0$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x\mathrm{e}^x - \mathrm{e}^x}{x^2} = \frac{\mathrm{e}^x(x-1)}{x^2}.$$

This is zero only at $x = 1$, so the only stationary point is $(1, \mathrm{e})$. The derivative is negative for $x < 1$ ($x \neq 0$) and positive for $x > 1$, so $(1, \mathrm{e})$ is a minimum.

The rest of the shape:

- For $x < 0$, $y < 0$ and y is decreasing. As $x \to -\infty$, $y \to 0^-$ (the exponential wins), so the negative x-axis is an asymptote. As $x \to 0^-$, $y \to -\infty$.
- For $x > 0$, $y > 0$. As $x \to 0^+$, $y \to +\infty$; y falls to the minimum value e at $x = 1$, then increases to $+\infty$, faster than any polynomial.
- The y-axis is a vertical asymptote and the curve never meets either axis.



Part (iii)(b)

Integrate $\dfrac{\mathrm{e}^x}{x^2}$ by parts, with $u = \mathrm{e}^x$ and $v = -x^{-1}$:

$$\int \frac{\mathrm{e}^x}{x^2}\,\mathrm{d}x = -\frac{\mathrm{e}^x}{x} + \int \frac{\mathrm{e}^x}{x}\,\mathrm{d}x, \qquad \text{so} \qquad \int \left(\frac{\mathrm{e}^x}{x} - \frac{\mathrm{e}^x}{x^2}\right)\mathrm{d}x = \frac{\mathrm{e}^x}{x} + c.$$

(Equivalently, $\frac{\mathrm{e}^x}{x} - \frac{\mathrm{e}^x}{x^2} = \frac{\mathrm{e}^x(x-1)}{x^2}$ is the derivative we found in (a).)

? **Ask yourself** Is it legitimate to integrate between a and $2a$ at all?

Both integrands blow up at $x = 0$, and near 0 the integral of e^x/x^2 diverges, so the interval must avoid 0. This needs $a \neq 0$; then a and $2a$ have the same sign, so 0 is never between them, and the integrals are finite.

The two integrals are equal exactly when their difference is zero:

$$\left[\frac{\mathrm{e}^x}{x}\right]_a^{2a} = \frac{\mathrm{e}^{2a}}{2a} - \frac{\mathrm{e}^a}{a} = \frac{\mathrm{e}^a}{2a}\left(\mathrm{e}^a - 2\right) = 0.$$

Since $\mathrm{e}^a/(2a) \neq 0$, we need $\mathrm{e}^a = 2$, so

$$a = \ln 2.$$

Part (iii)(c)

Exactly as in (b), for integers $m \neq n$ the two integrals can only be finite if 0 is not in the closed interval between m and n, so m and n are non-zero and of the same sign. Their difference is

$$\int_m^n \frac{\mathrm{e}^x}{x}\,\mathrm{d}x - \int_m^n \frac{\mathrm{e}^x}{x^2}\,\mathrm{d}x = \frac{\mathrm{e}^n}{n} - \frac{\mathrm{e}^m}{m},$$

so we need two distinct integers of the same sign at which $f(x) = \mathrm{e}^x/x$ takes the same value. In other words, a horizontal line must cut the graph from (a) at two integer values of x.

Ask yourself The curve for $x > 0$ is not one-to-one, since it comes down and goes back up. So why can't two positive integers give the same height?

Because the turning point is at $x = 1$, which is itself the smallest positive integer. Taking the cases in turn:

- *Both negative.* From (a), f is strictly decreasing on $x < 0$, so $m \neq n$ gives $f(m) \neq f(n)$.
- *Both positive.* Then $m, n \geq 1$, and $f'(x) > 0$ for $x > 1$, so f is strictly increasing on $[1, \infty)$. Again $m \neq n$ gives $f(m) \neq f(n)$.

A horizontal line can meet the positive branch twice, but one of the two meeting points then lies in $0 < x < 1$, where there are no integers. So no pair of distinct integers works, and it is impossible, as required.

Checks

- For (i), at $x = 1$ the antiderivative is $2\sqrt{2}$, and a numerical value of the integral over $[0, 1]$ is $2.8284\ldots$, matching.
- For (ii), the integral over $[1, 10]$ evaluates numerically to $1.4711\ldots$, the same as the change in $-(\sin x + x \cos x)/x^2$.
- For (iii)(b), with $a = \ln 2$ both integrals come to about 1.9224. Note also that $f(\ln 2) = 2/\ln 2 = f(2 \ln 2)$: the two limits sit at equal heights on either side of the minimum, as the sketch suggests they must.
- For (iii)(c), the closest any two integers come is $f(1) = \mathrm{e} \approx 2.718$ against $f(2) = \frac{1}{2}\mathrm{e}^2 \approx 3.695$.

Common slips

- In (b) and (c), forgetting that an interval containing 0 makes the integrals meaningless, and so missing why m and n must have the same sign.
- In (c), arguing "f is one-to-one" without restriction: it isn't (for example $f(\ln 2) = f(2 \ln 2)$). The argument needs the turning point at $x = 1$.

Beyond STEP

Parts (i) and (ii) are cases where neither piece of the integrand has an elementary antiderivative, yet the sum does. Liouville's theorem in differential algebra makes precise which functions have elementary antiderivatives; e^x/x, $\sin x/x$ and $\sqrt{1 + x^3}$ do not, which is why the cancellation is essential. Part (iii)(c) is an injectivity argument, and analysis courses make such reasoning about monotonicity and the mean value theorem rigorous; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2022, Question 2

Pure • Recurrence Relations

STEP II 2022, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Multiply the relation through by 2 and it becomes a statement about *second differences*: $u_{n+2} - 2u_{n+1} + u_n$. In (i) this is zero, so the first differences are constant and the sequence is linear. In (ii) and (iii) the second difference is a given constant or a given linear function of n, so the trick is to subtract off a polynomial with exactly that second difference, leaving a sequence of the type in (i).

Solution

Part (i)

Doubling the relation and rearranging gives, for all $n \geq 1$,

$$u_{n+2} - u_{n+1} = u_{n+1} - u_n.$$

So each difference equals the one before it, and by induction every difference equals the first one, $d = u_2 - u_1$. That is the constant difference between successive terms.

Adding up $n - 1$ of these differences,

$$u_n = u_1 + (n-1)d = dn + (u_1 - d),$$

which is a polynomial in n of degree 1 if $d \neq 0$ and of degree 0 (a constant) if $d = 0$. Either way the degree is at most 1.

Part (ii)

Put $v_n = t_n + pn^2$. Since

$$\tfrac{1}{2}\left(p(n+2)^2 + pn^2\right) = p(n^2 + 2n + 2) = p(n+1)^2 + p,$$

the relation for v_n becomes

$$t_{n+1} + p(n+1)^2 = \tfrac{1}{2}(t_{n+2} + t_n) + p(n+1)^2 + p - p,$$

that is, $t_{n+1} = \frac{1}{2}(t_{n+2} + t_n)$ for all $n \geq 1$. By part (i), $t_n = A + Bn$ for some constants A and B, so

$$v_n = pn^2 + Bn + A.$$

Because $p \neq 0$ the n^2 term survives, and v_n has degree exactly 2.

With $v_1 = v_2 = 0$:

$$p + B + A = 0, \qquad 4p + 2B + A = 0.$$

Subtracting, $3p + B = 0$, so $B = -3p$ and then $A = 2p$. Hence

$$v_n = p(n^2 - 3n + 2) = p(n-1)(n-2).$$

Part (iii)

This time no substitution is suggested, so we have to find one. Doubling the relation,

$$w_{n+2} - 2w_{n+1} + w_n = 2(an + b).$$

Ask yourself Which polynomial in n has a second difference that is linear in n?

Each difference lowers the degree of a polynomial by one, so we need a cubic. Write $\Delta^2 f(n) = f(n+2) - 2f(n+1) + f(n)$. Direct expansion gives

$$\Delta^2 n^3 = (n^3 + 6n^2 + 12n + 8) - 2(n^3 + 3n^2 + 3n + 1) + n^3 = 6n + 6,$$
$$\Delta^2 n^2 = (n^2 + 4n + 4) - 2(n^2 + 2n + 1) + n^2 = 2.$$

So $\Delta^2(cn^3 + en^2) = 6cn + 6c + 2e$, and this equals $2an + 2b$ for every n when

$$c = \frac{a}{3}, \qquad e = b - 3c = b - a.$$

Now put $w_n = t_n + \frac{a}{3}n^3 + (b-a)n^2$. Since Δ^2 is linear, $\Delta^2 w_n = \Delta^2 t_n + 2(an + b)$, and the relation becomes $\Delta^2 t_n = 0$, which is exactly the relation of part (i). So $t_n = A + Bn$ and

$$w_n = \frac{a}{3}n^3 + (b-a)n^2 + Bn + A.$$

Since $a \neq 0$, the cubic term is present and w_n has degree 3.

With $w_1 = w_2 = 0$:

$$\tfrac{a}{3} + (b-a) + B + A = 0,$$
$$\tfrac{8a}{3} + 4(b-a) + 2B + A = 0.$$

Subtracting the first from the second gives $\frac{7a}{3} + 3(b-a) + B = 0$, so $B = \frac{2a}{3} - 3b$, and then the first equation gives $A = 2b$. Therefore

$$w_n = \frac{a}{3}n^3 + (b-a)n^2 + \left(\frac{2a}{3} - 3b\right)n + 2b.$$

Since $w_1 = w_2 = 0$, the factors $n-1$ and $n-2$ must appear, and matching the n^3 and constant terms gives the tidier form

$$w_n = \frac{(n-1)(n-2)(an+3b)}{3}.$$

(Expanding $(n-1)(n-2)(an+3b)$ gives $an^3 + (3b-3a)n^2 + (2a-9b)n + 6b$, which agrees.)

Checks

- Setting $a = 0$, $b = p$ in (iii) should recover (ii), and it does: $(n-1)(n-2)\cdot 3p/3 = p(n-1)(n-2)$.
- Run the recurrence directly. With $a = 3$, $b = 0$ we have $w_{n+2} = 2w_{n+1} - w_n + 6n$, so from $w_1 = w_2 = 0$ we get $w_3 = 6$, $w_4 = 24$, $w_5 = 60$. The formula gives $(n-1)(n-2)n$, which is $6, 24, 60$.
- A degree-3 sequence has constant third differences. For $0, 0, 6, 24, 60$ the differences are $0, 6, 18, 36$, then $6, 12, 18$, then $6, 6$: constant, equal to $2a$.

Common slips

- Saying the sequence in (i) has degree exactly 1: if $u_1 = u_2$ it is constant.
- Trying only $w_n = t_n + kn^3$ in (iii). The n^3 term on its own produces $6kn + 6k$, so you also need an n^2 term to fix the constant.
- Forgetting to justify "degree exactly 2" (or 3) from $p \neq 0$ (or $a \neq 0$).

Beyond STEP

This is the discrete analogue of solving $y'' = f(x)$: the general solution is a particular solution plus the solution of the homogeneous equation, and here the homogeneous solutions are the linear sequences $A + Bn$. The same "complementary function plus particular solution" structure governs linear differential equations, and the method of guessing a polynomial particular solution carries over almost word for word. Boyce and DiPrima, *Elementary Differential Equations*, develops the continuous version.

WORKED SOLUTION

STEP II 2022, Question 3

Pure • Fibonacci Series and Decimals

STEP II 2022, Question 3

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

The recurrence $F_{r+1} - F_r - F_{r-1} = 0$ is the whole engine. Summing it against powers of $\frac{1}{10}$ and re-indexing turns each piece into a multiple of S_n plus a few boundary terms, which is exactly the generating-function trick. Part (i) is the safety net: it says the Fibonacci numbers grow no faster than powers of 2, so dividing by powers of 10 (or 100) gives tails that are tiny and easy to bound.

Solution

Part (i)

First, every $F_k \geq 0$ (by induction, since $F_0, F_1 \geq 0$ and sums of non-negative numbers are non-negative). Next, the sequence never decreases: $F_1 \geq F_0$, and for $k \geq 1$ we have $F_{k+1} - F_k = F_{k-1} \geq 0$.

So for every $k \geq 1$,

$$F_{k+1} = F_k + F_{k-1} \leq F_k + F_k = 2F_k.$$

Now fix $n \geq 1$ and induct on $r \geq n$. When $r = n$ the claim reads $F_n \leq F_n$. If $F_r \leq 2^{r-n}F_n$ for some $r \geq n$, then, since $r \geq 1$,

$$F_{r+1} \leq 2F_r \leq 2 \cdot 2^{r-n}F_n = 2^{(r+1)-n}F_n.$$

This completes the induction, so $F_r \leq 2^{r-n}F_n$ whenever $n \geq 1$ and $r \geq n$.

Part (ii)

We rewrite each of the three sums on the left in terms of S_n.

First sum. Put $k = r + 1$:

$$\sum_{r=1}^{n} \frac{F_{r+1}}{10^{r-1}} = 100 \sum_{k=2}^{n+1} \frac{F_k}{10^k} = 100\Big(S_n - \frac{F_1}{10} + \frac{F_{n+1}}{10^{n+1}}\Big)$$
$$= 100S_n - 10F_1 + \frac{F_{n+1}}{10^{n-1}}.$$

Second sum. This is simply $10S_n$.

Third sum. Put $k = r - 1$:

$$\sum_{r=1}^{n} \frac{F_{r-1}}{10^{r-1}} = \sum_{k=0}^{n-1} \frac{F_k}{10^k} = F_0 + S_n - \frac{F_n}{10^n}.$$

Subtracting the second and third sums from the first gives

$$(100 - 10 - 1)S_n - 10F_1 - F_0 + \frac{F_n}{10^n} + \frac{F_{n+1}}{10^{n-1}} = 89S_n - 10F_1 - F_0 + \frac{F_n}{10^n} + \frac{F_{n+1}}{10^{n-1}},$$

which is the required result.

Part (iii)

Combine the three sums on the left of (ii) into one. Its rth term is $(F_{r+1} - F_r - F_{r-1})/10^{r-1} = 0$, so the left-hand side is zero. With $F_0 = 0$ and $F_1 = 1$, this gives

$$89S_n = 10 - \frac{F_n}{10^n} - \frac{F_{n+1}}{10^{n-1}}. \qquad (*)$$

? **Ask yourself** How do we know the two leftover terms vanish as $n \to \infty$?

This is where part (i) earns its keep. Taking $n = 1$ there gives $F_r \le 2^{r-1}$ for all $r \ge 1$, so

$$0 \le \frac{F_n}{10^n} \le \frac{1}{2}\Big(\frac{1}{5}\Big)^n, \qquad 0 \le \frac{F_{n+1}}{10^{n-1}} \le \frac{2^n}{10^{n-1}} = 10\Big(\frac{1}{5}\Big)^n,$$

and both bounds tend to 0. Letting $n \to \infty$ in $(*)$, we get $89S_n \to 10$, that is,

$$\sum_{r=1}^{\infty} \frac{F_r}{10^r} = \frac{10}{89}.$$

For the tail, apply part (i) with $n = 7$ (note $F_7 = 13$): for $r \ge 7$ we have $F_r \le 2^{r-7} \cdot 13$, so

$$\sum_{r=7}^{\infty} \frac{F_r}{10^r} \le \frac{13}{10^7} \sum_{j=0}^{\infty} \frac{2^j}{10^j} = \frac{13}{10^7} \cdot \frac{1}{1 - \frac{1}{5}} = \frac{65}{4} \times 10^{-7} = 1.625 \times 10^{-6} < 2 \times 10^{-6}.$$

Digits of $\frac{1}{89}$. The first six terms add up to

$$S_6 = 0.1 + 0.01 + 0.002 + 0.0003 + 0.00005 + 0.000008 = 0.112358.$$

The tail $T = \frac{10}{89} - S_6$ is strictly positive (its terms are positive) and less than 2×10^{-6}, so

$$0.112358 < \frac{10}{89} < 0.112360, \qquad \text{hence} \qquad 0.0112358 < \frac{1}{89} < 0.0112360.$$

Every number strictly between these bounds begins $0.011235\ldots$, so the first six digits after the decimal point of $\frac{1}{89}$ are $0, 1, 1, 2, 3, 5$. The strict lower bound matters: it rules out the expansion being $0.0112357\ldots$.

Part (iv)

Read the target in pairs of digits after the leading 00:

$$0.00\ 01\ 01\ 02\ 03\ 05\ 08\ 13\ 21\ 34\ 55\ldots$$

These pairs are $F_1, F_2, \ldots, F_{10}$, each written as two digits. So the number should be $\frac{1}{100}U$, where

$$U = \sum_{r=1}^{\infty} \frac{F_r}{100^r}.$$

Summing U. The series converges absolutely, because $F_r \leq 2^{r-1}$ by part (i). We can therefore shift and combine convergent series term by term, just as in (ii):

$$U - \frac{U}{100} - \frac{U}{100^2} = \frac{F_1}{100} + \sum_{r=2}^{\infty} \frac{F_r - F_{r-1} - F_{r-2}}{100^r} = \frac{1}{100}.$$

Here the $r = 1$ term is $F_1/100$ (since $F_0 = 0$), and every later bracket is zero. So $U \cdot \frac{9899}{10000} = \frac{1}{100}$, giving $U = \frac{100}{9899}$. The candidate is therefore

$$\frac{U}{100} = \frac{1}{9899},$$

with $r = 1$ and $s = 9899$, both positive integers less than 10000.

Checking the digits. Let U_{10} be the sum of the first ten terms. Because each of $F_1, \ldots, F_{10}$ is less than 100, the terms do not overlap, and

$$U_{10} = 0.01010203050813213455$$

exactly, with 20 decimal places. By part (i) with $n = 11$ (and $F_{11} = 89$), the remaining terms satisfy

$$0 < U - U_{10} \leq \frac{89}{100^{11}} \sum_{j=0}^{\infty} \Big(\frac{2}{100}\Big)^j = \frac{89}{10^{22}} \cdot \frac{100}{98} < 10^{-20}.$$

So $U_{10} < U < U_{10} + 10^{-20}$, and the first 20 decimal digits of U are those of U_{10}. Dividing by 100 shifts these digits two places to the right:

$$\frac{1}{9899} = 0.0001010203050813213455\ldots,$$

which is exactly the required start.

Checks

- Long division gives $1/89 = 0.011235955\ldots$. The digits 011235 agree, and the next digits $9, 5, 5$ show the Fibonacci terms "carrying" into each other once they reach two digits.
- The tail beyond $r = 6$ in (iii) is actually $10/89 - 0.112358 \approx 1.55 \times 10^{-6}$, safely below our bound of 1.625×10^{-6}.
- A computer check of $1/9899$ to 22 places gives 0.0001010203050813213455. The next pair is 90, not $F_{11} = 89$, because $F_{12} = 144$ overflows its two-digit slot and carries 1 into the one before.

Common slips

- Getting the boundary terms wrong when re-indexing in (ii). Writing out the first and last terms of each shifted sum avoids this.
- In (iii), not showing that $F_n/10^n$ and $F_{n+1}/10^{n-1}$ tend to zero. Part (i) is there precisely to do this.
- Forgetting that the tail is *strictly* positive. An upper bound alone does not fix the sixth digit.

Beyond STEP

The function $G(x) = \sum F_r x^r$ is the *generating function* of the Fibonacci numbers, and the shifting argument shows that $G(x) = x/(1 - x - x^2)$ for $|x| < \frac{1}{2}$. Putting in $x = \frac{1}{10}$ and $x = \frac{1}{100}$ gives $\frac{10}{89}$ and $\frac{100}{9899}$. Generating functions turn recurrences into algebra and are a central tool in combinatorics. Herbert Wilf's *generatingfunctionology* is a lively introduction. Making "rearranging convergent series" rigorous is a first-year analysis topic: see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2022, Question 4

Pure • Modulus Functions

STEP II 2022, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

A modulus sign only matters at the point where its contents change sign, so on each interval between such points every modulus can be replaced by plus or minus its contents. Going the other way, a term $A|x-p|$ puts a corner at $x=p$ where the gradient jumps by $2A$. So to build a single formula, we match the gradient jumps (and, in the last part, the jumps in the x^2 coefficient) and then fix the constant.

Solution

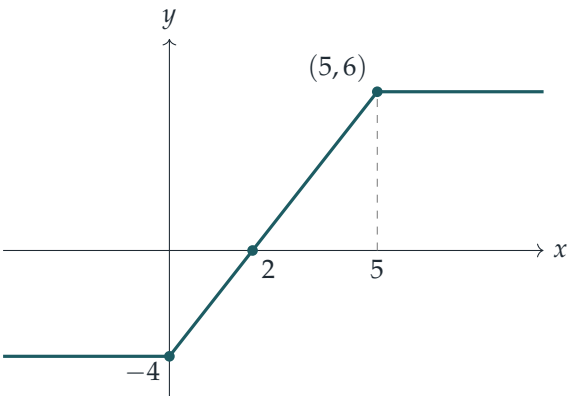
Part (i)

The critical points are $x=0$ (where $|x|$ switches) and $x=5$ (where $|x-5|$ switches).

- For $x \leq 0$: $|x| = -x$ and, since $x-5<0$, $|x-5| = 5-x$. So $\mathrm{f}(x) = -x-(5-x)+1 = -4$.
- For $0 \leq x \leq 5$: $|x| = x$ and $|x-5| = 5-x$. So $\mathrm{f}(x) = x-(5-x)+1 = 2x-4$.
- For $x \geq 5$: $|x| = x$ and $|x-5| = x-5$. So $\mathrm{f}(x) = x-(x-5)+1 = 6$.

At $x=0$ and $x=5$ both relevant expressions are valid (each modulus is zero there or has a definite sign), so the overlapping endpoints are consistent: $-4 = 2(0)-4$ and $2(5)-4 = 6$. This is the required piecewise form.

The graph is horizontal at height -4, rises with gradient 2 from $(0,-4)$ through $(2,0)$ to $(5,6)$, then is horizontal at height 6.



Part (ii)

Part (i) suggests trying

$$g(x) = A|x| + B|x-5| + Cx + D.$$

Passing from left to right through $x = 0$, the term $A|x|$ changes from $-Ax$ to Ax, so the gradient jumps by $2A$; similarly $B|x-5|$ makes the gradient jump by $2B$ at $x = 5$. The given g has gradients -1, 3, 1 on the three pieces, so

$$2A = 3 - (-1) = 4, \qquad 2B = 1 - 3 = -2,$$

giving $A = 2$, $B = -1$. For $x \geq 5$ the formula is $2x - (x-5) + Cx + D = (1+C)x + 5 + D$, and this must be $x + 10$, so $C = 0$ and $D = 5$. Hence

$$\boxed{g(x) = 2|x| - |x-5| + 5.}$$

Checking the other two pieces: for $x \leq 0$, $-2x - (5-x) + 5 = -x$; for $0 \leq x \leq 5$, $2x - (5-x) + 5 = 3x$. Both agree with the definition of g.

Ask yourself Why did a formula of this shape have any chance of working?

Each piece of g is linear and the pieces join up continuously (0 at $x = 0$ and 15 at $x = 5$ from both sides). A sum of moduli of linear terms plus a linear function is always continuous and piecewise linear, with corners only where the moduli switch. If g had a jump, no such formula could produce it.

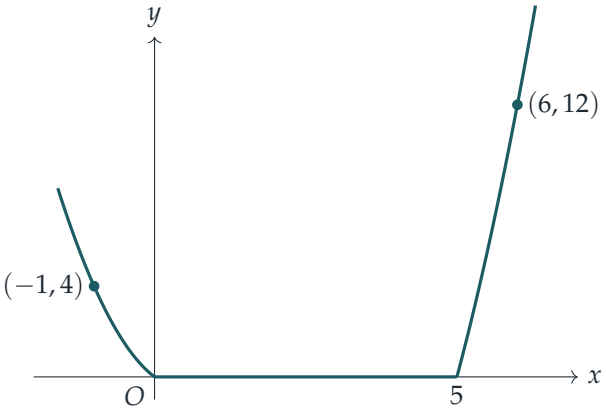
Part (iii)

Now the switching points are $x = 0$ (for $|x|$) and the roots of $x(x-5)$, which are again 0 and 5. The product $x(x-5)$ is positive for $x < 0$ and $x > 5$, and negative between.

- For $x \leq 0$: $h(x) = x^2 - x + 4x + (x^2 - 5x) = 2x^2 - 2x = 2x(x-1)$.
- For $0 \leq x \leq 5$: $h(x) = x^2 - x - 4x - (x^2 - 5x) = 0$.

- For $x \geq 5$: $\mathrm{h}(x) = x^2 - x - 4x + (x^2 - 5x) = 2x^2 - 10x = 2x(x-5)$.

So h is identically zero on $[0,5]$. To the left it is part of the parabola $y = 2x(x-1)$: it meets the origin with gradient $4x - 2 = -2$, and it increases without bound as $x \to -\infty$ (for example $\mathrm{h}(-1) = 4$). To the right it is part of $y = 2x(x-5)$: it leaves $(5,0)$ with gradient $4x - 10 = 10$ (for example $\mathrm{h}(6) = 12$). The two arms are *not* mirror images: the right arm leaves the axis five times as steeply. The graph is continuous, with corners at $(0,0)$ and $(5,0)$, and $\mathrm{h}(x) \geq 0$ everywhere.



Part (iv)

Part (iii) shows the tool we need: $|x(x-5)|$ equals $x^2 - 5x$ outside $[0,5]$ and $-(x^2-5x)$ inside, so it can switch an x^2 term on and off. The function k has x^2 coefficient 0 outside $[0,5]$ and 2 inside. In $ax^2 + d|x(x-5)|$ the coefficient of x^2 is $a+d$ outside and $a-d$ inside, so we need $a+d=0$ and $a-d=2$, that is, $a=1$ and $d=-1$. Thus $x^2 - |x(x-5)|$ has the right quadratic behaviour, and

$$x^2 - |x(x-5)| = \begin{cases} 5x & x \leq 0 \text{ or } x \geq 5, \\ 2x^2 - 5x & 0 \leq x \leq 5. \end{cases}$$

Subtracting this from k leaves a piecewise linear remainder:

$$\mathrm{k}(x) - \left(x^2 - |x(x-5)|\right) = \begin{cases} 5x & x \leq 5, \\ 50 - 5x & x \geq 5. \end{cases}$$

This has a single corner at $x = 5$, where the gradient jumps from 5 to -5, a jump of -10; so it is $-5|x-5| + (\text{linear})$. For $x \leq 5$, $-5|x-5| = 5x - 25$, so we add 25 to get $5x$. This suggests

$$\boxed{\mathrm{k}(x) = x^2 - |x(x-5)| - 5|x-5| + 25.}$$

Verification. We substitute on each interval.

- For $x \leq 0$: $x(x-5) \geq 0$ and $x - 5 < 0$, so the formula gives $x^2 - (x^2 - 5x) - 5(5 - x) + 25 = 5x + 5x = 10x$.

- For $0 \leq x \leq 5$: $x(x-5) \leq 0$ and $x-5 \leq 0$, so it gives $x^2 + (x^2 - 5x) - 5(5-x) + 25 = 2x^2$.
- For $x \geq 5$: $x(x-5) \geq 0$ and $x-5 \geq 0$, so it gives $x^2 - (x^2 - 5x) - 5(x-5) + 25 = 50$.

These are exactly the three pieces of k, so the formula is correct for every real x.

The single formula isn't unique. For example, $|x(x-5)| = |x|\,|x-5|$, so the answer can be rewritten, and other combinations (such as ones using $x|x|$) also work. Any answer is fine if it passes the three-interval check above.

Checks

- Spot values: $g(-2) = 2$ from both the definition and $2(2) - 7 + 5$; $g(7) = 17 = 14 - 2 + 5$. For k: $k(3) = 18$ and $9 - |-6| - 10 + 25 = 18$; $k(-1) = -10$ and $1 - 6 - 30 + 25 = -10$.
- Continuity is a quick sanity test: every single formula built from moduli is continuous, and indeed g and k agree at the joins ($k(5) = 50 = 2 \cdot 5^2$).
- All four formulas were compared with their piecewise forms by computer at 10^5 random points in $[-20, 25]$, with agreement to rounding error.

Common slips

- Sketching h with symmetric arms. The left arm is $2x(x-1)$ and the right arm is $2x(x-5)$, with gradients -2 and 10 at the corners.
- Forgetting that $|x(x-5)|$ changes sign at *two* points, so in (iii) it flips in the middle interval, not just at 0.
- Writing down a formula for k without substituting into all three intervals; the verification is required.

Beyond STEP

The gradient-jump bookkeeping here is a first taste of a general fact: every continuous piecewise linear function on $\mathbb{R}$ with finitely many corners is a linear function plus a combination of terms $|x-p|$. This is the idea behind linear splines in numerical analysis and the "ReLU" units of neural networks, since $\max(x, 0) = \frac{1}{2}(x + |x|)$. For a careful treatment of continuity and functions defined piece by piece, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2022, Question 5

Pure • Inequalities in a Triangle

STEP II 2022, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Splitting the triangle into three pieces from P shows that, however P moves, the weighted sum $ax + by + cz$ stays fixed at 2Δ. Each later part is then an optimisation under that single linear constraint. Part (i) compares $x + y + z$ with the fixed quantity, the identity in (iii)(a) does the same for $x^2 + y^2 + z^2$ (it is the Cauchy–Schwarz inequality in disguise), and for the maximum we use (i) the other way round, with c in place of a.

Solution

Part (i)

Since $a > b$ and $y \geq 0$, we have $by \leq ay$; since $a > c$ and $z \geq 0$, we have $cz \leq az$. Adding ax to both sides of the sum of these gives

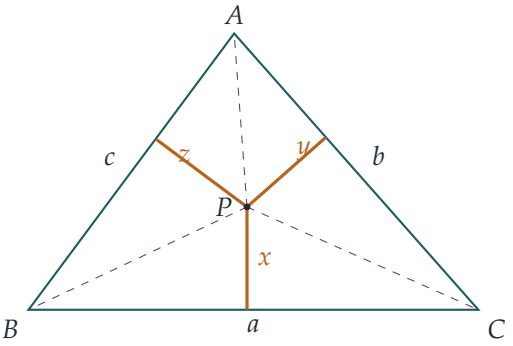
$$ax + by + cz \leq ax + ay + az = a(x + y + z),$$

as required. Because $a - b > 0$ and $a - c > 0$, equality holds exactly when $(a - b)y + (a - c)z = 0$, that is, when $y = z = 0$. We will need this equality case.

Part (ii)(a)

Joining P to the vertices splits ABC into triangles PBC, PCA and PAB (some may be degenerate if P is on a side). Triangle PBC has base $BC = a$ and height x, and so on, so

$$\Delta = \tfrac{1}{2}ax + \tfrac{1}{2}by + \tfrac{1}{2}cz, \qquad \text{that is,} \qquad ax + by + cz = 2\Delta.$$



Conversely, any $x, y, z \geq 0$ with $ax + by + cz = 2\Delta$ are the distances from exactly one point P in the triangle (or on its sides): the three triangle areas $\frac{1}{2}ax$, $\frac{1}{2}by$, $\frac{1}{2}cz$ are non-negative and add up to Δ, and they pin down P (they are its areal coordinates). So in what follows we can treat the problem as purely algebraic: optimise over all non-negative x, y, z with $ax + by + cz = 2\Delta$.

Part (ii)(b)

By part (i) and then (ii)(a),

$$x + y + z \geq \frac{ax + by + cz}{a} = \frac{2\Delta}{a}.$$

Equality holds exactly when $y = z = 0$, and then $ax = 2\Delta$. So the minimum value of the sum is

$$x + y + z = \frac{2\Delta}{a}, \qquad \text{attained at} \quad x = \frac{2\Delta}{a},\ y = 0,\ z = 0.$$

Geometrically, $y = z = 0$ puts P on both CA and AB, so $P = A$, and x is the altitude from A to the longest side BC.

Part (iii)(a)

Expanding the three squared differences,

$$(bx - ay)^2 + (cy - bz)^2 + (az - cx)^2 = b^2x^2 + a^2y^2 + c^2y^2 + b^2z^2 + a^2z^2 + c^2x^2 - 2(abxy + bcyz + cazx),$$

while

$$(ax + by + cz)^2 = a^2x^2 + b^2y^2 + c^2z^2 + 2(abxy + bcyz + cazx).$$

Adding, the cross terms cancel and every product of one of a^2, b^2, c^2 with one of x^2, y^2, z^2 appears exactly once, giving $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$. This proves the identity for all real values.

Part (iii)(b)

Put $ax + by + cz = 2\Delta$ into the identity:

$$(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (bx - ay)^2 + (cy - bz)^2 + (az - cx)^2 + 4\Delta^2 \geq 4\Delta^2,$$

since squares are non-negative. Hence

$$x^2+y^2+z^2 \geq \frac{4\Delta^2}{a^2+b^2+c^2}.$$

Ask yourself A lower bound is only the minimum if it is actually reached. Can all three squares vanish at a genuine point P?

Equality needs $bx = ay$, $cy = bz$ and $az = cx$. These say (x, y, z) is a multiple of (a, b, c): if $x = \lambda a$, then $y = \frac{b}{a}x = \lambda b$ and $z = \frac{c}{a}x = \lambda c$, and all three equations hold. The constraint gives $\lambda(a^2+b^2+c^2) = 2\Delta$, so $\lambda = \dfrac{2\Delta}{a^2+b^2+c^2} > 0$. Then x, y, z are all positive and satisfy $ax+by+cz = 2\Delta$, so by the remark in (ii)(a) they belong to a real point P (strictly inside the triangle). So the minimum value of the sum of squares is

$$\frac{4\Delta^2}{a^2+b^2+c^2}, \qquad \text{attained at} \quad x = \frac{2a\Delta}{a^2+b^2+c^2},\ y = \frac{2b\Delta}{a^2+b^2+c^2},\ z = \frac{2c\Delta}{a^2+b^2+c^2}.$$

Part (iv)

Ask yourself Part (i) used that a is the largest side. What happens if we use the smallest side c instead?

Since $a > c$, $b > c$ and $x, y \geq 0$, the argument of part (i) reversed gives $ax+by+cz \geq c(x+y+z)$, with equality exactly when $x = y = 0$. So

$$x+y+z \leq \frac{2\Delta}{c}.$$

Also, as $x, y, z \geq 0$,

$$x^2+y^2+z^2 = (x+y+z)^2 - 2(xy+yz+zx) \leq (x+y+z)^2,$$

with equality exactly when $xy+yz+zx = 0$. Putting these together (squaring is safe because both sides are non-negative),

$$x^2+y^2+z^2 \leq (x+y+z)^2 \leq \frac{4\Delta^2}{c^2}.$$

Both inequalities become equalities when $x = y = 0$: then $xy+yz+zx = 0$ and $cz = 2\Delta$. So the maximum value of the sum of squares is

$$\frac{4\Delta^2}{c^2}, \qquad \text{attained at} \quad x = 0,\ y = 0,\ z = \frac{2\Delta}{c},$$

which is $P = C$, the vertex opposite the shortest side. (The second inequality is an equality only when $x = y = 0$, so this is the only point where the maximum occurs.)

Checks

Take the acute triangle with sides $a = 7$, $b = 6$, $c = 5$. Heron's formula gives $\Delta = \sqrt{9 \cdot 2 \cdot 3 \cdot 4} = 6\sqrt{6} \approx 14.70$. The answers are then: minimum of $x + y + z$ is $12\sqrt{6}/7 \approx 4.20$; minimum of $x^2 + y^2 + z^2$ is $864/110 = 432/55 \approx 7.85$; maximum is $864/25 = 34.56$. The order makes sense: at $P = C$ the sum of squares is just the altitude from C squared, $(12\sqrt{6}/5)^2 = 34.56$. A computer check over many random acute triangles, with thousands of random points P in each, confirmed $ax + by + cz = 2\Delta$ and all three bounds, with the extremes reached at A, at the point with distances proportional to (a, b, c), and at C.

Common slips

- Giving a lower bound without showing it is attained. In (iii)(b) you must check that the equality point has non-negative distances satisfying the constraint, so that it is a genuine P.
- Mixing up the vertices: the minimum of $x + y + z$ is at A (opposite the longest side), but the maximum of $x^2 + y^2 + z^2$ is at C (opposite the shortest side).
- Dropping the condition $x, y, z \geq 0$ in (iv); without it, $x^2 + y^2 + z^2 \leq (x + y + z)^2$ is false.

Beyond STEP

The identity in (iii)(a) is Lagrange's identity, and it proves the Cauchy–Schwarz inequality $(\mathbf{u} \cdot \mathbf{v})^2 \leq |\mathbf{u}|^2 |\mathbf{v}|^2$ in three dimensions: the minimum point is where (x, y, z) is parallel to (a, b, c), the foot of the perpendicular from the origin to the plane $ax + by + cz = 2\Delta$. The maximum of a convex function such as $x^2 + y^2 + z^2$ over a triangle is always at a corner, a first glimpse of linear and convex optimisation. The point found in (iii)(b) is the symmedian (or Lemoine) point of the triangle. For more on dot products and projections, see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP II 2022, Question 6

Pure • Orthogonal Trajectories

STEP II 2022, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

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The key idea

Two curves meet at right angles exactly when the product of their gradients there is -1. So for a whole family of curves we want a gradient formula that holds at a point (x, y) *whichever* member of the family passes through it. That means eliminating the parameter (a, b, c or k) between the curve's equation and its derivative, leaving a differential equation in x, y and $\frac{\mathrm{d}y}{\mathrm{d}x}$ alone. Then the perpendicular family is found by replacing the gradient m by $-1/m$ and solving.

Solution

Throughout, $x > 0$ and $y > 0$.

Part (i)(a)

Differentiating $x^2 + y^2 = 2ax$ implicitly with respect to x gives

$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 2a.$$

This still contains a, which changes from curve to curve. The curve's equation gives $2a = (x^2 + y^2)/x$, so substituting and multiplying by x,

$$2x^2 + 2xy\frac{\mathrm{d}y}{\mathrm{d}x} = x^2 + y^2 \quad \Longrightarrow \quad 2xy\frac{\mathrm{d}y}{\mathrm{d}x} = y^2 - x^2,$$

as required.

For the second family, differentiating $x^2 + y^2 = 2by$ gives (remembering that the right-hand side also depends on x through y)

$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 2b\frac{\mathrm{d}y}{\mathrm{d}x}.$$

Now $2b = (x^2 + y^2)/y$; substituting and multiplying by y,

$$2xy + 2y^2\frac{\mathrm{d}y}{\mathrm{d}x} = (x^2 + y^2)\frac{\mathrm{d}y}{\mathrm{d}x} \quad \Longrightarrow \quad (x^2 - y^2)\frac{\mathrm{d}y}{\mathrm{d}x} = 2xy.$$

Part (i)(b)

Suppose a curve of each family passes through (x, y), with $y \neq x$. Since $x, y > 0$ we have $2xy \neq 0$ and, because $y \neq x$ and both are positive, $x^2 - y^2 \neq 0$. So we may divide in both differential equations: the two gradients are

$$m_1 = \frac{y^2 - x^2}{2xy}, \qquad m_2 = \frac{2xy}{x^2 - y^2}, \qquad \text{so} \qquad m_1 m_2 = \frac{y^2 - x^2}{x^2 - y^2} = -1.$$

Both gradients are finite and their product is -1, so the tangents are perpendicular.

Ask yourself What goes wrong on the line $y = x$, and does that mean the result fails there?

On $y = x$ the second equation reads $0 \cdot \frac{\mathrm{d}y}{\mathrm{d}x} = 2x^2 \neq 0$, which no finite gradient satisfies: the tangent to the second curve is vertical. That breaks the algebra, not the geometry, so we look directly.

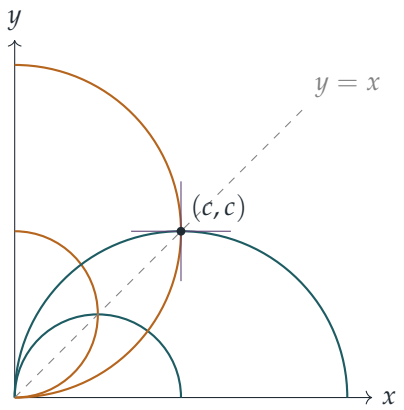
If (c, c) lies on $x^2 + y^2 = 2ax$ then $2c^2 = 2ac$, so $a = c$; similarly $b = c$. The first curve is the circle with centre $(c, 0)$ and radius c, and (c, c) is its highest point, so the tangent there is horizontal (indeed $m_1 = 0$ when $y = x$):

$$y = c.$$

The second curve is the circle with centre $(0, c)$ and radius c, and (c, c) is its right-most point, so its tangent is vertical:

$$x = c.$$

A horizontal and a vertical line are perpendicular, so **yes**: the tangents are perpendicular at every meeting point, including those on $y = x$.



Circles of the first family (teal) touch the y-axis at the origin; those of the second (amber) touch the x-axis there.

Part (ii)

For $y = c\ln x$ we have $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{c}{x}$. To eliminate c, note that at a point of the first quadrant on such a curve, $y > 0$ forces $\ln x \neq 0$, so $c = y/\ln x$ and

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y}{x\ln x}.$$

This gradient is never zero, so a perpendicular curve through the same point must have gradient $-x\ln x/y$. The second family therefore satisfies

$$y\frac{\mathrm{d}y}{\mathrm{d}x} = -x\ln x.$$

Separating variables, and integrating $x\ln x$ by parts with $u = \ln x$, $\mathrm{d}v = x\,\mathrm{d}x$,

$$\int y\,\mathrm{d}y = -\int x\ln x\,\mathrm{d}x \quad\Longrightarrow\quad \tfrac{1}{2}y^2 = -\tfrac{1}{2}x^2\ln x + \tfrac{1}{4}x^2 + C.$$

Multiplying by 4, the second family is

$$2y^2 + 2x^2\ln x - x^2 = k,$$

one curve for each constant k (only the parts with $x > 0$, $y > 0$ count). Conversely, differentiating this implicitly gives $4y\frac{\mathrm{d}y}{\mathrm{d}x} + 4x\ln x + 2x - 2x = 0$, which is exactly the equation above, so wherever one of these curves meets some $y = c\ln x$ the product of the gradients is $\frac{y}{x\ln x}\cdot\frac{-x\ln x}{y} = -1$.

Part (iii)

Differentiating $y^2 = 4k(x+k)$ gives $2y\frac{\mathrm{d}y}{\mathrm{d}x} = 4k$, so, as $y > 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2k}{y}.$$

Ask yourself Here there is only one family. Where does the "second curve" come from?

Fix a point (x, y) in the first quadrant and ask which members of the family pass through it. The value of k must satisfy

$$4k^2 + 4xk - y^2 = 0,$$

a quadratic in k. Its discriminant $16x^2 + 16y^2$ is positive, so there are two distinct real roots k_1, k_2, and their product is

$$k_1k_2 = -\frac{y^2}{4}.$$

(This is negative, so both roots are non-zero, as the family requires.) A quadratic has at most two roots, so if two different curves of the family meet at (x, y), their parameters must be exactly k_1 and k_2. Their gradients there are $2k_1/y$ and $2k_2/y$, with product

$$\frac{4k_1k_2}{y^2} = \frac{4}{y^2} \cdot \left(-\frac{y^2}{4}\right) = -1.$$

So the tangents are perpendicular at every such point, as required. (These curves are parabolas with focus at the origin; the family is its own set of orthogonal trajectories.)

Checks

- In (i), take $(x, y) = (1, 2)$: then $a = \frac{5}{2}$, $b = \frac{5}{4}$, and the gradients are $m_1 = \frac{3}{4}$ and $m_2 = -\frac{4}{3}$, with product -1.
- In (ii), at $(e, 1)$ the log curve has $c = 1$ and gradient $1/e$; the formula for the second family gives gradient $-e \cdot 1/1 = -e$. Again the product is -1.
- In (iii), at $(0, 2)$ the quadratic is $4k^2 - 4 = 0$, so $k = \pm 1$: the parabolas $y^2 = 4(x + 1)$ and $y^2 = -4(x - 1)$, with gradients 1 and -1 there.
- All four results were also confirmed by computer at a few thousand random points, using numerical derivatives of the curves themselves.

Common slips

- Forgetting the factor $\frac{\mathrm{d}y}{\mathrm{d}x}$ on the right when differentiating $2by$.
- In (ii), writing the perpendicular gradient as $-x/c$ and integrating with c still present: c belongs to the first curve, so it must be eliminated before solving.
- Dividing by $x^2 - y^2$ in (i)(b) without saying why it is non-zero, and so missing that the line $y = x$ needs separate treatment.

Beyond STEP

Families of curves meeting everywhere at right angles are called *orthogonal trajectories*. They appear in physics as field lines and equipotentials, and in complex analysis: the real and imaginary parts of an analytic function always have perpendicular level curves, which is why the circles in (i) arise from $1/z$ and the self-orthogonal parabolas in (iii) from $\sqrt{z}$. Boyce and DiPrima, *Elementary Differential Equations*, treats orthogonal trajectories, and Needham, *Visual Complex Analysis*, explains the complex-analysis picture beautifully.

WORKED SOLUTION

STEP II 2022, Question 7

Pure • Roots of Complex Polynomials

STEP II 2022, Question 7

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www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

The equation $\mathrm{h}(w) = 0$ splits naturally as $w^5(nw+1) = -(w+n)$, and taking moduli turns it into a statement about $|w|$ alone. Once every root is known to lie on the unit circle, each non-real root pairs with its conjugate, which is also its reciprocal. That gives the quadratic factors, and after that it's just comparing coefficients and a sign argument.

Solution

Part (i)

(a) Group the terms of $\mathrm{h}(w) = 0$ as

$$w^5(nw+1) = -(w+n).$$

Neither bracket can vanish. If $w = -1/n$ then $\mathrm{h}(w) = n - \frac{1}{n} \neq 0$, and if $w = -n$ then $\mathrm{h}(w) = n^5(n^2-1) \neq 0$, since $n \geq 2$. Taking moduli and squaring,

$$|w^5|^2\,|nw+1|^2 = |w+n|^2.$$

For any complex u and real c we have $|u+c|^2 = (u+c)(\bar{u}+c) = |u|^2 + 2c\,\mathrm{Re}(u) + c^2$. So

$$|w+n|^2 = |w|^2 + 2n\,\mathrm{Re}(w) + n^2 = \mathrm{f}(w),$$
$$|nw+1|^2 = n^2|w|^2 + 2n\,\mathrm{Re}(w) + 1 = \mathrm{g}(w).$$

Both are positive, so $|w^5|^2 = \mathrm{f}(w)/\mathrm{g}(w)$. Taking the positive square root gives $|w^5| = \sqrt{\mathrm{f}(w)/\mathrm{g}(w)}$, as required.

(b) The $\mathrm{Re}(w)$ terms cancel in the difference:

$$\mathrm{f}(w) - \mathrm{g}(w) = n^2 + |w|^2 - n^2|w|^2 - 1 = (n^2-1)\left(1 - |w|^2\right).$$

Suppose, for contradiction, that $|w| < 1$. Since $n^2 - 1 > 0$, we get $\mathrm{f}(w) > \mathrm{g}(w) > 0$, so $\mathrm{f}(w)/\mathrm{g}(w) > 1$. By (a), $|w|^5 > 1$, so $|w| > 1$, which contradicts $|w| < 1$. Hence $|w| \geq 1$.

(c) The same argument works the other way round. If $|w| > 1$, then $\mathrm{f}(w) < \mathrm{g}(w)$, so $|w|^5 < 1$ and $|w| < 1$, a contradiction. So $|w| \leq 1$ as well, and together with (b), $|w| = 1$.

Ask yourself Is there a slicker reason why $|w| \leq 1$?

Yes. The coefficients $n, 1, 0, 0, 0, 1, n$ read the same backwards, so $z^6\,\mathrm{h}(1/z) = \mathrm{h}(z)$. Hence if w is a root, so is $1/w$, and applying (b) to $1/w$ gives $|w| \leq 1$.

Part (ii)

(a) Since h has real coefficients, its non-real roots come in conjugate pairs. The six roots are distinct and non-real, so they form three pairs $w_k,\ \bar{w}_k$ $(k = 1, 2, 3)$. By part (i), $|w_k| = 1$, so $w_k\bar{w}_k = 1$, and

$$(z - w_k)(z - \bar{w}_k) = z^2 - (w_k + \bar{w}_k)z + w_k\bar{w}_k = z^2 - a_k z + 1,$$

where $a_k = 2\,\mathrm{Re}(w_k)$ is real. The polynomial h has degree 6 and leading coefficient n, so

$$\mathrm{h}(z) = n(z^2 - a_1 z + 1)(z^2 - a_2 z + 1)(z^2 - a_3 z + 1).$$

(b) The z^5 terms of the product arise by taking $-a_k z$ from one factor and z^2 from the other two, so the coefficient of z^5 is $-n(a_1 + a_2 + a_3)$. In h that coefficient is 1, so

$$a_1 + a_2 + a_3 = -\frac{1}{n}.$$

(c) First multiply two of the factors:

$$(z^2 - a_1 z + 1)(z^2 - a_2 z + 1) = z^4 - (a_1 + a_2)z^3 + (2 + a_1 a_2)z^2 - (a_1 + a_2)z + 1.$$

Multiplying by $z^2 - a_3 z + 1$, the z^3 terms come from $z^3 \times 1$, $z^2 \times (-a_3 z)$ and $z \times z^2$:

$$-(a_1 + a_2) - a_3(2 + a_1 a_2) - (a_1 + a_2) = -2(a_1 + a_2 + a_3) - a_1 a_2 a_3.$$

There is no z^3 term in h, so $a_1 a_2 a_3 = -2(a_1 + a_2 + a_3)$, and by (b),

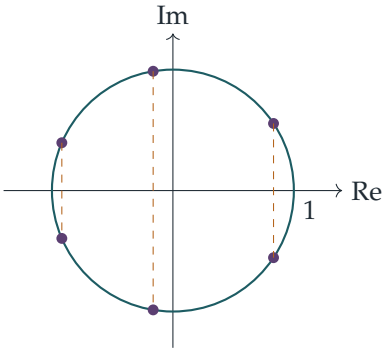
$$a_1 a_2 a_3 = \frac{2}{n}.$$

(d)

Ask yourself How do the signs of the a_k tell us about real parts?

Each conjugate pair $w_k, \bar{w}_k$ shares the real part $\mathrm{Re}(w_k) = a_k/2$. So the number of roots with negative real part is twice the number of negative a_k.

Since $a_1 a_2 a_3 = 2/n > 0$, none of the a_k is zero, and an even number of them are negative: none or two. They can't all be positive, because $a_1 + a_2 + a_3 = -1/n < 0$. So exactly two of the a_k are negative and one is positive, and **four** of the six roots have negative real part.



The figure shows the case $n = 2$. The six roots lie on the unit circle in three conjugate pairs, and two of the pairs are to the left of the imaginary axis.

Checks

- As a sanity check that no root is real, note that $\mathrm{h}(1) = 2n+2$ and $\mathrm{h}(-1) = 2n-2$ are both non-zero. Since the roots lie on the unit circle, ± 1 are the only real candidates.
- For $n = 2$, solving numerically gives $a_k \approx -1.8368,\ -0.3272,\ 1.6640$. These sum to $-0.5 = -\frac{1}{2}$, their product is $1.000 = \frac{2}{2}$, and two are negative, matching (b), (c) and (d). A computer check of $n = 2, \ldots, 29$ confirmed that every root has modulus 1 and that four roots have negative real part each time.

Common slips

- Dividing by $nw + 1$ without checking that $w = -1/n$ isn't a root.
- In (c), missing one of the three ways to make z^3, especially the term $-2a_3$ that comes from the constant 2 in $2 + a_1 a_2$.
- In (d), counting negative a_k but forgetting that each one accounts for *two* roots.

Beyond STEP

A polynomial whose coefficients read the same backwards is called *palindromic* (or self-reciprocal). Its roots come in pairs $w, 1/w$, and the substitution $u = z + 1/z$ halves its degree. Here $a_k = w_k + 1/w_k$ is exactly that u. Questions about when all the roots lie on the unit circle lead to results such as the Schur–Cohn test and to Rouché's theorem in complex analysis. Needham's *Visual Complex Analysis* is a beautiful place to see these ideas geometrically.

WORKED SOLUTION

STEP II 2022, Question 8

Pure • Invariant Lines of Matrices

STEP II 2022, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

A line through the origin is invariant when every point on it is sent to a point on the same line, so it is enough to see where one direction vector goes. Writing the line as $y = mx$ and asking that $\mathbf{M}\binom{1}{m}$ be a multiple of $\binom{1}{m}$ turns everything into a quadratic in m. After that, parts (ii) and (iii) are about the sum and product of its roots, and the vertical line, which has no gradient, needs a separate look every time.

Solution

Part (i)

Take the line $y = mx$, with direction $\binom{1}{m}$. Its image is

$$\mathbf{M}\begin{pmatrix}1\\m\end{pmatrix} = \begin{pmatrix}a+bm\\c+dm\end{pmatrix}.$$

Because $\mathbf{M}$ is linear, the line is invariant exactly when this image lies on $y = mx$, that is when

$$c + dm = m(a + bm) \quad \Longleftrightarrow \quad bm^2 + (a-d)m - c = 0.$$

So every non-vertical invariant line has a gradient that satisfies this equation, and every root gives an invariant line.

Before calling the gradients "the roots" we must check that this really is a quadratic. The y-axis has direction $\binom{0}{1}$, whose image is $\binom{b}{d}$; this lies on the y-axis exactly when $b = 0$. So if neither invariant line is the y-axis, then $b \neq 0$. The equation is then a genuine quadratic, with at most two roots. Both invariant lines are non-vertical with different gradients, so their gradients are two distinct roots. That means they are exactly the two roots, as required.

If one invariant line is the y-axis, then $b = 0$ and the equation collapses to $(a-d)m - c =$

0. Since $a \neq d$ (this is where that condition is used), the other invariant line has gradient

$$m = \frac{c}{a-d}.$$

Part (ii)

Case $b \neq 0$. Let the gradients be m_1 and m_2. From part (i),

$$m_1 + m_2 = -\frac{a-d}{b}, \qquad m_1 m_2 = -\frac{c}{b}.$$

If the lines make angles α_1 and α_2 with the x-axis, then

$$\tan(\alpha_1 - \alpha_2) = \frac{m_1 - m_2}{1 + m_1 m_2}.$$

An angle of 45° between the lines means this equals ± 1, so $(m_1 - m_2)^2 = (1 + m_1 m_2)^2$. Writing the left side as $(m_1 + m_2)^2 - 4m_1 m_2$ gives

$$\frac{(a-d)^2}{b^2} + \frac{4c}{b} = \left(1 - \frac{c}{b}\right)^2.$$

Multiplying by b^2 gives $(a-d)^2 + 4bc = (b-c)^2$, that is,

$$(a-d)^2 = (b-c)^2 - 4bc.$$

Case $b = 0$. One line is the y-axis and the other has gradient $c/(a-d)$. For it to make 45° with the y-axis it must have gradient ± 1, so $c = \pm(a-d)$ and $(a-d)^2 = c^2$. With $b = 0$ the right-hand side of the required result is also c^2, so the result holds in this case too.

Part (iii)

? **Ask yourself** What does "equal angles with $y = x$" mean in terms of gradients?

Two different lines through the origin make the same angle with $y = x$ exactly when one is the mirror image of the other in $y = x$. Reflecting in $y = x$ swaps the coordinates, so it sends $y = mx$ to $x = my$, which is $y = \frac{1}{m}x$ when $m \neq 0$. It sends the x-axis to the y-axis and the y-axis to the x-axis.

Case $b \neq 0$. Neither line is vertical. If one had gradient 0, its mirror image would be the y-axis, which is not invariant, so the lines would not be symmetric. Otherwise the condition is $m_2 = 1/m_1$, that is, $m_1 m_2 = 1$. By part (i) this is $-c/b = 1$, that is, $b + c = 0$.

Case $b = 0$. One line is the y-axis, so the other must be the x-axis: $c/(a-d) = 0$, that is, $c = 0$. Given $b = 0$, this is again the condition $b + c = 0$.

So in all cases the necessary and sufficient condition is

$$b + c = 0, \quad \text{that is,} \quad c = -b.$$

(When the matrix is symmetric, so $b = c$, the invariant lines are perpendicular instead. Here the off-diagonal entries are opposite.)

Part (iv)

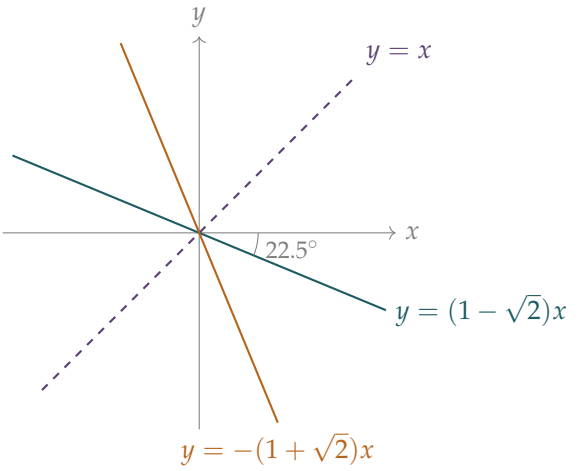
Put $c = -b$ into the condition of part (ii):

$$(a-d)^2 = (2b)^2 + 4b^2 = 8b^2.$$

We need $b \neq 0$: if $b = c = 0$, then $(a-d)^2 = 0$, which contradicts $a \neq d$. Take $b = 1$, $c = -1$ and $a - d = 2\sqrt{2}$, for example $a = \sqrt{2}$, $d = -\sqrt{2}$:

$$\mathbf{M} = \begin{pmatrix} \sqrt{2} & 1 \\ -1 & -\sqrt{2} \end{pmatrix}.$$

The quadratic of part (i) is $m^2 + 2\sqrt{2}\,m + 1 = 0$, with distinct real roots $m = -\sqrt{2} \pm 1$. So this matrix has exactly two invariant lines, as the question assumes. Their product is 1, so they are mirror images in $y = x$. Their angles with the x-axis are $-22.5°$ and $-67.5°$, since $\tan 22.5° = \sqrt{2} - 1$ and $\tan 67.5° = \sqrt{2} + 1$. They therefore meet at $45°$, and each makes $67.5°$ with $y = x$.



Checks

- For the example in part (iv), $(a-d)^2 = 8$ and $(b-c)^2 - 4bc = 4 + 4 = 8$, so the condition of part (ii) holds. We can also check one line directly: $\mathbf{M}\begin{pmatrix} 1 \\ 1-\sqrt{2} \end{pmatrix} = \begin{pmatrix} \sqrt{2}+1-\sqrt{2} \\ -1-\sqrt{2}+2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1-\sqrt{2} \end{pmatrix}$, so every point of $y = (1-\sqrt{2})x$ is in fact fixed.
- A quick special case for part (i): for $\mathbf{M} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ the invariant lines are the axes. The formula from the y-axis case gives the other gradient as $0/(2-1) = 0$, which is correct.

- Parts (i)–(iii) were confirmed by computer on many thousands of random matrices: the eigenvector directions satisfy the quadratic, pairs at 45° satisfy the identity, and the lines are symmetric about $y = x$ exactly when $b + c = 0$.

Common slips

- Forgetting the vertical line. The quadratic only sees gradients, so the case $b = 0$ has to be handled separately in every part.
- In part (iii), writing $m_1 = -m_2$. That is symmetry about the *axes*. Symmetry about $y = x$ needs $m_1 m_2 = 1$.
- Using $\tan\theta = 1$ without the $\pm$. Squaring $(m_1 - m_2)^2 = (1 + m_1 m_2)^2$ deals with both signs at once.

Beyond STEP

The invariant lines through the origin are the *eigenvector* directions of $\mathbf{M}$, and the quadratic in part (i) is the eigenvector equation $\mathbf{Mv} = \lambda\mathbf{v}$ with λ eliminated. A real symmetric matrix always has perpendicular eigenvectors (the spectral theorem). This question shows how antisymmetric off-diagonal entries instead make the eigenvectors symmetric about $y = x$. Strang's *Introduction to Linear Algebra* develops eigenvectors and the spectral theorem.

WORKED SOLUTION

STEP II 2022, Question 9

Mechanics • Friction and Moments

STEP II 2022, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

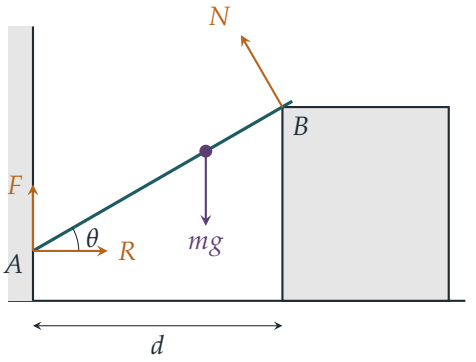
The plank is light, so only three things push on it: the point mass, the reaction at the prism's edge, and the wall. Taking moments about the point where the plank touches the wall removes both wall forces at once and gives the reaction at the edge straight away. Resolving vertically then gives the friction, and every later part is the friction inequality $|F| \leq \mu R$, rearranged.

Solution

Model. The plank is a light rigid rod, the mass is a particle of weight mg, the edge of the prism is smooth, and the wall is rough with coefficient μ. Let A be the point where the plank touches the wall and B the point where it rests on the edge. Since B is a horizontal distance d from the wall,

$$AB = d \sec\theta.$$

At B the plank rests on an edge, so the smooth reaction N is perpendicular to the *plank*. At A the wall gives a horizontal normal reaction R and a vertical friction force F, which we take as positive upwards (a negative F means it acts downwards).



Moments about A. Both wall forces pass through A. The weight acts at distance x along the plank, so its moment arm is $x\cos\theta$; N is perpendicular to the plank at distance $d\sec\theta$:

$$N\,d\sec\theta = mg\,x\cos\theta \quad\Longrightarrow\quad N = \frac{mg\,x\cos^2\theta}{d}.$$

Resolving. The direction of N makes an angle θ with the vertical, tilted towards the wall. Horizontally and vertically,

$$R = N\sin\theta = \frac{mg\,x\cos^2\theta\sin\theta}{d},$$
$$F = mg - N\cos\theta = mg\left(1 - \frac{x\cos^3\theta}{d}\right).$$

Note that $R > 0$, so the plank really does press on the wall.

Part (i)

If $x = d\sec^3\theta$ then $x\cos^3\theta/d = 1$, so $F = 0$: no friction acts at the wall.

Part (ii)

Equilibrium with friction requires $|F| \le \mu R$.

If $x > d\sec^3\theta$, then $x\cos^3\theta/d > 1$, so $F < 0$ (friction acts downwards; without it the end A would slide up the wall). The condition $-F \le \mu R$ reads

$$mg\left(\frac{x\cos^3\theta}{d} - 1\right) \le \mu\,\frac{mg\,x\cos^2\theta\sin\theta}{d}.$$

Multiplying by the positive quantity $d\sec^3\theta/(mg)$ gives

$$x - d\sec^3\theta \le \mu\,x\tan\theta, \qquad \text{so} \qquad \mu \ge \frac{x - d\sec^3\theta}{x\tan\theta},$$

dividing by $x\tan\theta > 0$ (as θ is acute). This is the required result.

If $x < d\sec^3\theta$, then $F > 0$ and the condition $F \le \mu R$ becomes, in the same way,

$$\mu \ge \frac{d\sec^3\theta - x}{x\tan\theta}.$$

Ask yourself Which way does friction act, and how do we know without guessing?

We don't need to guess. Taking F positive upwards, the vertical equation tells us its sign. Friction can always be written as $|F| \le \mu R$, and the sign of F just decides which of the two inequalities we get.

Part (iii)

Both cases of (ii) say $|x - d\sec^3\theta| \le \mu x \tan\theta$, which is the same as the pair

$$x - d\sec^3\theta \le \mu x\tan\theta \qquad \text{and} \qquad d\sec^3\theta - x \le \mu x\tan\theta.$$

The second gives $d\sec^3\theta \le x(1+\mu\tan\theta)$. Since $1+\mu\tan\theta > 0$,

$$\frac{x}{d} \ge \frac{\sec^3\theta}{1+\mu\tan\theta}.$$

The first gives $x(1-\mu\tan\theta) \le d\sec^3\theta$. If $\mu < \cot\theta$ then $\mu\tan\theta < 1$, so $1-\mu\tan\theta > 0$ and we may divide without reversing the inequality:

$$\frac{x}{d} \le \frac{\sec^3\theta}{1-\mu\tan\theta}.$$

(If $\mu \ge \cot\theta$ there is no upper bound: the friction can hold the mass however far up the plank it sits.)

Part (iv)

The mass lies between the wall and the edge exactly when $x < AB = d\sec\theta$. Combining with the lower bound from (iii),

$$\frac{d\sec^3\theta}{1+\mu\tan\theta} \le x < d\sec\theta.$$

Dividing by $d\sec\theta > 0$ and multiplying by $1+\mu\tan\theta > 0$,

$$\sec^2\theta < 1+\mu\tan\theta \quad\Longleftrightarrow\quad \tan^2\theta < \mu\tan\theta,$$

using $\sec^2\theta = 1+\tan^2\theta$. Since $\tan\theta > 0$ we can divide by it, giving $\tan\theta < \mu$, as required.

Checks

- Take $\theta = 45^\circ$, $d = 1$ and $\mu = 0.5$. Then $\sec^3\theta = 2\sqrt{2} \approx 2.83$, and (iii) allows $1.89 \le x \le 5.66$. At $x = 2$, say, $F = mg(1 - 2/2\sqrt{2}) \approx 0.29mg$ and $N = mg$, $R = mg/\sqrt{2}$ and $\mu R \approx 0.35mg$, so equilibrium holds. Here the mass is beyond the edge ($AB = \sqrt{2}$), consistent with (iv), since $\tan 45^\circ = 1 > \mu$.
- Limiting case $\mu = 0$ (smooth wall): the bounds in (iii) collapse to $x = d\sec^3\theta$, the only position from (i).
- A computer check over many random values of θ, d, μ and x confirmed that the forces balance (including moments about other points), and that $|F| \le \mu R$ holds exactly when the inequalities of (iii) hold.

Common slips

- Taking the reaction at the edge as vertical. The plank rests on an edge, so the smooth reaction is perpendicular to the plank, not to the top of the prism.
- Using d instead of $d\sec\theta$ for the distance AB in the moments equation.
- Dividing $x(1-\mu\tan\theta) \leq d\sec^3\theta$ without checking the sign of $1-\mu\tan\theta$; that is why the condition $\mu < \cot\theta$ appears.

Beyond STEP

This plank is *statically determinate*: three unknown reactions and three equilibrium equations, so friction is pinned down exactly. Add one more support and the equations no longer fix the forces; engineers then bring in the elasticity of the beam to decide how the load is shared. The bounds in (iii) are a small example of a "region of equilibrium", a theme that returns in the study of stability. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully.

WORKED SOLUTION

STEP II 2022, Question 10

Mechanics • Projectiles on Inclined Plane

STEP II 2022, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Part (i) is the standard trajectory equation, and the trick is to read it as a *quadratic in* $\tan\alpha$, using $\sec^2\alpha = 1 + \tan^2\alpha$. A target point can be hit exactly when that quadratic has a real root, so the whole question reduces to a discriminant condition. Written in the horizontal coordinates (x, y), that condition turns out to describe a disc, and every later part is a matter of reading off where a line meets a circle.

Solution

Throughout we model the projectile as a particle moving freely under gravity, with no air resistance, and take $0 < \theta < \frac{\pi}{2}$.

Part (i)

Take the horizontal distance and the height measured from the point of projection. At time t,

$$s = ut\cos\alpha, \qquad h = ut\sin\alpha - \tfrac{1}{2}gt^2.$$

The horizontal motion gives $t = \dfrac{s}{u\cos\alpha}$, and substituting,

$$h = s\tan\alpha - \frac{gs^2}{2u^2\cos^2\alpha},$$

as required.

Part (ii)

Let $P = (x, y, y\tan\theta)$ and write $s = \sqrt{x^2 + y^2}$ for its horizontal distance from the cannon. Since the cannon can point in any direction, we aim it in the vertical plane through P; the motion then takes place in that plane, and by part (i) we need an angle

α (possibly negative, since the cannon can point downwards) with

$$y\tan\theta = s\tan\alpha - \frac{gs^2}{2u^2}\left(1+\tan^2\alpha\right).$$

If $s=0$ then P is the origin, which is reached by firing straight up, and the inequality below holds there because $\sec^2\theta > \tan^2\theta$. For $s>0$, put $T=\tan\alpha$, which can be any real number. The condition is that

$$\frac{gs^2}{2u^2}T^2 - sT + \left(y\tan\theta + \frac{gs^2}{2u^2}\right) = 0$$

has a real root T, that is, its discriminant is non-negative:

$$s^2 \geq 4\cdot\frac{gs^2}{2u^2}\left(y\tan\theta+\frac{gs^2}{2u^2}\right) \iff u^4 \geq 2gu^2y\tan\theta + g^2s^2,$$

after dividing by $s^2>0$ and multiplying by u^4. Dividing by g^2 and using $s^2=x^2+y^2$,

$$x^2+y^2+\frac{2u^2\tan\theta}{g}y \leq \frac{u^4}{g^2}.$$

Completing the square in y, and using $1+\tan^2\theta=\sec^2\theta$,

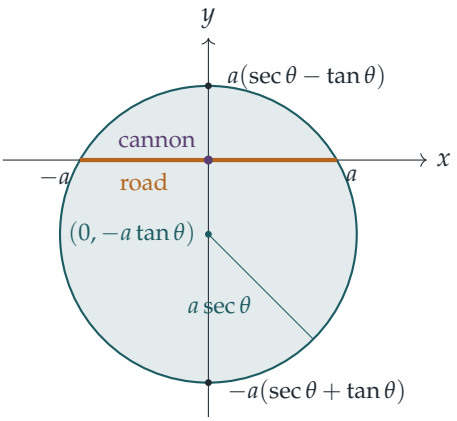
$$x^2+\left(y+\frac{u^2\tan\theta}{g}\right)^2 \leq \frac{u^4}{g^2}\left(1+\tan^2\theta\right) = \frac{u^4\sec^2\theta}{g^2},$$

which is the required condition.

Ask yourself Could the projectile hit the sloping ground *before* it reaches P?

No. In the vertical plane of the shot, the height of the projectile minus the height of the ground is a quadratic in the horizontal distance with negative leading coefficient (the parabola minus a straight line), and it vanishes at the cannon and at P. So it is positive in between: the projectile stays above the ground until it lands at P.

From now on write $a = u^2/g$. The reachable points are those whose horizontal projection lies in the disc with centre $(0,-a\tan\theta)$ and radius $a\sec\theta$.



The figure (drawn with $\theta = 30°$) shows the reachable region seen from above; "up the plane" is the positive y direction.

Part (iii)

Directly up the plane means $x = 0$, $y > 0$. The condition becomes $(y + a\tan\theta)^2 \leq a^2\sec^2\theta$, and since $y + a\tan\theta > 0$ this is

$$y \leq a(\sec\theta - \tan\theta).$$

A point on the plane with horizontal coordinate y lies at distance $y\sec\theta$ from the origin, measured along the plane. So the furthest point up the plane is at distance

$$a\sec\theta(\sec\theta - \tan\theta) = a\,\frac{1-\sin\theta}{\cos^2\theta} = a\,\frac{1-\sin\theta}{(1-\sin\theta)(1+\sin\theta)} = \frac{u^2}{g(1+\sin\theta)},$$

as required.

Directly down the plane means $x = 0$, $y < 0$. Now the condition gives $y \geq -a(\sec\theta + \tan\theta)$, so the furthest point down the plane is at distance

$$a\sec\theta(\sec\theta + \tan\theta) = a\,\frac{1+\sin\theta}{\cos^2\theta} = \frac{u^2}{g(1-\sin\theta)}$$

from the cannon.

Part (iv)

The road is the line $y = 0$ (which lies in the plane, at height 0). Putting $y = 0$ in the condition gives

$$x^2 \leq a^2\sec^2\theta - a^2\tan^2\theta = a^2,$$

so the reachable road is $-a \leq x \leq a$, a length of $\dfrac{2u^2}{g}$, the same as on level ground.

Now place the cannon at $(0, c, c\tan\theta)$. Its distance from the road (the x-axis) is $\sqrt{c^2 + c^2\tan^2\theta} = |c|\sec\theta$, so $r = |c|\sec\theta$. Everything in part (ii) depends only on positions *relative to the cannon*, and the plane looks the same from every point on it, so we can use the condition with y replaced by $y - c$. The road ($y = 0$) is reached where

$$x^2 \leq a^2\sec^2\theta - (a\tan\theta - c)^2.$$

? **Ask yourself** Should the cannon be placed above the road or below it?

The reachable length is $2\sqrt{a^2\sec^2\theta - (a\tan\theta - c)^2}$, which is largest when the subtracted square is zero, that is when $c = a\tan\theta > 0$. So the cannon goes *up* the plane from the road, exactly far enough that the road passes through the centre of the reachable disc, and it then cuts a diameter. The best distance is

$$r = a\tan\theta\sec\theta = \frac{u^2\sin\theta}{g\cos^2\theta},$$

and the maximum length of road is the diameter of the disc,

$$2a\sec\theta = \frac{2u^2}{g\cos\theta}.$$

Checks

- With $\theta = 0$ (level ground) part (ii) becomes $x^2 + y^2 \leq (u^2/g)^2$, the familiar maximum range u^2/g in every direction; both answers in (iii) reduce to u^2/g, and in (iv) $r = 0$ with length $2u^2/g$.
- The up-slope answer $u^2/\big(g(1+\sin\theta)\big)$ is the standard maximum range up an inclined plane, a good sign that the disc is right.
- With $u = 12\ \mathrm{m\,s^{-1}}$, $g = 9.8\ \mathrm{m\,s^{-2}}$ and $\theta = 0.5$ rad, a computer scan over all firing angles found the maximum range up the slope to be 9.932 m and down the slope 28.226 m, matching $u^2/\big(g(1\pm\sin\theta)\big)$; the best road length with the moved cannon was 33.49 m $= 2u^2/(g\cos\theta)$.

Common slips

- Treating (x, y) as distances along the plane. They are horizontal coordinates, so a distance up the plane is $y\sec\theta$, not y.
- Forgetting that the cannon may point below the horizontal: $\tan\alpha$ ranges over all real numbers, which is why the discriminant condition is the whole story.
- Placing the moved cannon below the road. The disc's centre is down the slope from the cannon, so the cannon must sit above the road.

Beyond STEP

The disc in part (ii) is a slice of the *envelope* of all possible trajectories, the "paraboloid of safety" $z = \dfrac{u^2}{2g} - \dfrac{g(x^2+y^2)}{2u^2}$; intersecting this paraboloid with any plane gives a conic, here a circle in projection. Envelopes of families of curves are a natural first-year topic in both mechanics and differential equations. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see projectile problems treated with this kind of geometric care.

WORKED SOLUTION

STEP II 2022, Question 11

Probability and Statistics • Expected Loss Optimisation

STEP II 2022, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Split the story by what happens to the T tested sticks. If any of them is infected, nothing goes on the network and the loss is zero; only when all T are clean does the network gamble come into play. Writing the expected loss as a function of q turns (ii) into a one-variable maximisation, and comparing consecutive values of T (a difference, not a derivative, since T is an integer) handles (iii).

Solution

Throughout, a loss of $-x$ means a gain of £x. Let L be the net loss in pounds. Sticks are infected independently, each clean with probability q, so any particular set of m sticks is entirely clean with probability q^m. Which T sticks are chosen doesn't matter, by symmetry.

Part (i)

Take $T > 0$ first. There are three outcomes.

- Some tested stick is infected (probability $1 - q^T$): the rest are destroyed, nothing is used, and $L = 0$.
- All T tested sticks are clean, but at least one of the other $N - T$ is infected (probability $q^T(1 - q^{N-T})$): we pay for disinfection, so $L = D$.
- All N sticks are clean (probability $q^T q^{N-T} = q^N$): we gain £1 on each of the $N - T$ sticks used, so $L = -(N - T)$.

Hence

$$\begin{aligned} \mathrm{E}(L) &= D\,q^T\left(1 - q^{N-T}\right) - (N - T)\,q^N \\ &= D\,q^T - (N - T + D)\,q^N. \end{aligned}$$

When $T = 0$ all N sticks are used: $L = D$ with probability $1 - q^N$ and $L = -N$ with probability q^N, giving $D - (N + D)q^N$. That is the same formula with $T = 0$, so for every $0 \leq T < N$

$$\mathrm{E}(L) = D\,q^T - (N - T + D)\,q^N.$$

Part (ii)

Bounds on α. Since $D > 0$, $T \geq 0$ and $N - T > 0$, numerator and denominator satisfy $DT \geq 0$ and $N(N - T + D) > 0$, so $\alpha \geq 0$. Also

$$N(N - T + D) - DT = N(N - T) + D(N - T) = (N - T)(N + D) > 0,$$

so $DT < N(N - T + D)$ and $\alpha < 1$.

The maximum. Write $f(q) = Dq^T - (N - T + D)q^N$ for $0 \leq q \leq 1$, with N, D, T fixed.

If $T = 0$ then $\alpha = 0$ and $f(q) = D - (N + D)q^N$, which decreases in q, so the greatest value is at $q = 0$, i.e. where $q^N = 0 = \alpha$.

If $T \geq 1$, differentiate:

$$f'(q) = DTq^{T-1} - N(N - T + D)q^{N-1} = N(N - T + D)\,q^{T-1}\big(\alpha - q^{N-T}\big).$$

Ask yourself How do we know this stationary point is a maximum, and the greatest value on the whole interval?

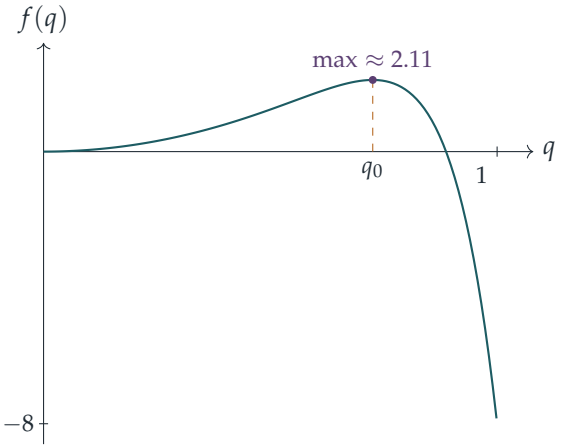
The factor $N(N - T + D)q^{T-1}$ is positive for $0 < q \leq 1$. Because $N - T \geq 1$, q^{N-T} is strictly increasing from 0 to 1, and $0 < \alpha < 1$, so there is exactly one $q_0 \in (0, 1)$ with $q_0^{N-T} = \alpha$. For $q < q_0$ we have $f'(q) > 0$ and for $q > q_0$ we have $f'(q) < 0$. So f rises up to q_0 and falls afterwards: its greatest value on $[0, 1]$ occurs where $q^{N-T} = \alpha$, as required.

The greatest value. At q_0 we have $q_0^{N-T} = \alpha$, so $q_0 = \alpha^{1/(N-T)}$ and $q_0^T = \alpha^{T/(N-T)} = \alpha^k$. Then

$$\begin{aligned} f(q_0) &= q_0^T\big(D - (N - T + D)\,q_0^{N-T}\big) = \alpha^k\Big(D - (N - T + D)\cdot\frac{DT}{N(N - T + D)}\Big) \\ &= \alpha^k\Big(D - \frac{DT}{N}\Big) = \frac{D(N - T)\,\alpha^k}{N}. \end{aligned}$$

So the greatest expected net loss is £$D(N - T)\alpha^k/N$. (For $T = 0$ this reads £D, matching $f(0) = D$ if we take $\alpha^0 = 1$.)

The graph shows f for $N = 10$, $T = 2$, $D = 5$, where $\alpha = \frac{1}{13}$.



Part (iii)

Now fix N, D and q, and write $E_T = Dq^T - (N - T + D)q^N$. Since T only takes integer values, "increasing in T" means $E_{T+1} > E_T$ for $0 \leq T \leq N - 2$. Subtracting,

$$\begin{aligned} E_{T+1} - E_T &= Dq^{T+1} - Dq^T + q^N \\ &= q^T(q^{N-T} - Dp), \end{aligned}$$

using $q - 1 = -p$.

Ask yourself The bracket depends on T. How can one β work for every T at once?

Because $0 < q \leq 1$ and $N - T \leq N$, we have $q^{N-T} \geq q^N$, so it is enough to make $q^N > Dp$, a condition that doesn't involve T. By Bernoulli's inequality (an easy induction: if $(1-p)^n \geq 1 - np$ then $(1-p)^{n+1} \geq (1-np)(1-p) = 1 - (n+1)p + np^2$),

$$q^N = (1-p)^N \geq 1 - Np.$$

So if $1 - Np > Dp$, that is $p < \dfrac{1}{N+D}$, then $q^{N-T} \geq q^N \geq 1 - Np > Dp$, and also $q^T > 0$. Therefore, with

$$\beta = \frac{1}{N+D} > 0,$$

every $p < \beta$ gives $E_{T+1} > E_T$ for all T, so the expected loss is an increasing function of T.

For such p the smallest of $E_0, E_1, \ldots, E_{N-1}$ is E_0. So when p is small enough, the best policy is $T = 0$: test nothing and use all N sticks. This makes sense: if infections are rare, testing mostly destroys good sticks that would have earned money.

Checks

- With $N = 1$, $T = 0$ the formula gives $D - (1 + D)q$: lose D with probability p and gain 1 with probability q, as expected.
- $q = 1$ (no infection at all) gives $D - (N - T + D) = -(N - T)$, a sure gain of $N - T$.
- For $N = 10$, $T = 2$, $D = 5$: $\alpha = \frac{1}{13}$, $k = \frac{1}{4}$, and the maximum is $4 \cdot 13^{-1/4} \approx 2.107$, matching the peak of the graph above. A computer check confirmed the part (i) formula by listing every infection pattern for $N \leq 7$.
- For $N = 10$, $D = 5$, $p = 0.3$ (well above $\beta = \frac{1}{15}$), the expected loss actually *decreases* with T, from about 4.58 at $T = 0$ to 0.03 at $T = 9$: so some condition on p is genuinely needed.

Common slips

- Forgetting the outcome "a tested stick is infected", which contributes 0 to the expectation but still uses up probability $1 - q^T$.
- Counting the gain as £N instead of £$(N - T)$: the tested sticks are destroyed even when they are clean.
- In (iii), differentiating with respect to the integer T without comment, or choosing a β that depends on T.

Beyond STEP

This is a small problem in statistical decision theory: choose an action (here T) to minimise expected loss when a parameter (p) is unknown. Part (ii) is a *minimax* idea: it finds the worst case over all possible p for each policy, which is what a cautious manager might compare. The same trade-off between the cost of testing and the cost of missed defects lies behind acceptance sampling and group testing. Blitzstein and Hwang, *Introduction to Probability*, is a good place to build the expectation toolkit used here.

WORKED SOLUTION

STEP II 2022, Question 12

Probability and Statistics • Mean, Median and Mode

STEP II 2022, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Everything flows from the cumulative distribution function F, which here is a simple polynomial. Since F is increasing on $[0,1]$, a number a lies below the median exactly when $\mathrm{F}(a) < \frac{1}{2}$. So to compare the mean or the mode with the median, we never need to find the median itself: we just evaluate F at the mean or the mode and compare with $\frac{1}{2}$.

Solution

Part (i)

Since the total probability is 1,

$$1 = k\int_0^1 \left(x^n - x^{n+1}\right)\mathrm{d}x = k\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{k}{(n+1)(n+2)},$$

so $k = (n+1)(n+2)$, as required. Then

$$\mu = k\int_0^1 \left(x^{n+1} - x^{n+2}\right)\mathrm{d}x = (n+1)(n+2)\cdot\frac{1}{(n+2)(n+3)} = \frac{n+1}{n+3}.$$

Part (ii)

For $0 \le x \le 1$,

$$\mathrm{F}(x) = (n+1)(n+2)\left(\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2}\right) = x^{n+1}\big((n+2) - (n+1)x\big).$$

Because $\mathrm{f} > 0$ on $(0,1)$, F is strictly increasing there, and the median m is the unique point with $\mathrm{F}(m) = \frac{1}{2}$. Hence

$$\mu < m \iff \mathrm{F}(\mu) < \tfrac{1}{2}.$$

Now with $\mu = \frac{n+1}{n+3}$,

$$(n+2) - (n+1)\mu = \frac{(n+2)(n+3) - (n+1)^2}{n+3} = \frac{3n+5}{n+3},$$

so

$$\mathrm{F}(\mu) = \left(\frac{n+1}{n+3}\right)^{n+1} \frac{3n+5}{n+3}.$$

The condition $\mathrm{F}(\mu) < \frac{1}{2}$ therefore becomes, after multiplying by the positive quantity $2\left(\frac{n+3}{n+1}\right)^{n+1}$,

$$\frac{2(3n+5)}{n+3} < \left(\frac{n+3}{n+1}\right)^{n+1}.$$

Since $\frac{2(3n+5)}{n+3} = \frac{6(n+3)-8}{n+3} = 6 - \frac{8}{n+3}$ and $\frac{n+3}{n+1} = 1 + \frac{2}{n+1}$, this is exactly the given inequality. So μ is less than the median whenever that inequality holds.

To prove the inequality, write $N = n+1$, so $N \geq 3$ is an integer. Every term of the binomial expansion of $\left(1 + \frac{2}{N}\right)^N$ is positive, and there are $N+1 \geq 4$ terms, so the sum is at least its first four terms:

$$\begin{aligned}\left(1 + \frac{2}{N}\right)^N &\geq 1 + N \cdot \frac{2}{N} + \frac{N(N-1)}{2} \cdot \frac{4}{N^2} + \frac{N(N-1)(N-2)}{6} \cdot \frac{8}{N^3} \\ &= 3 + 2\left(1 - \frac{1}{N}\right) + \frac{4}{3}\left(1 - \frac{3}{N} + \frac{2}{N^2}\right) \\ &= \frac{19}{3} - \frac{6}{N} + \frac{8}{3N^2}.\end{aligned}$$

Ask yourself The four-term sum is only a lower bound. How do we show it beats $6 - \frac{8}{N+2}$?

We subtract and clear the positive denominator $3N^2(N+2)$:

$$\begin{aligned}&3N^2(N+2)\left[\frac{19}{3} - \frac{6}{N} + \frac{8}{3N^2} - 6 + \frac{8}{N+2}\right] \\ &\quad = N^2(N+2) - 18N(N+2) + 8(N+2) + 24N^2 \\ &\quad = N^3 + 8N^2 - 28N + 16.\end{aligned}$$

For $N \geq 3$ we have $N^2 + 8N - 28 \geq 9 + 24 - 28 = 5 > 0$, so $N^3 + 8N^2 - 28N + 16 = N(N^2 + 8N - 28) + 16 > 0$. Hence

$$6 - \frac{8}{n+3} < \frac{19}{3} - \frac{6}{N} + \frac{8}{3N^2} \leq \left(1 + \frac{2}{n+1}\right)^{n+1},$$

the given inequality holds for every integer $n > 1$, and so the median of X is greater than μ.

Part (iii)

On $(0,1)$,

$$\mathrm{f}'(x) = k\left(nx^{n-1} - (n+1)x^n\right) = kx^{n-1}\big(n - (n+1)x\big),$$

which is positive for $x < \frac{n}{n+1}$ and negative for $x > \frac{n}{n+1}$. So f rises then falls, and the mode is $M = \frac{n}{n+1}$.

As in part (ii), $M > m$ if and only if $\mathrm{F}(M) > \frac{1}{2}$. Since $(n+2) - (n+1)M = (n+2) - n = 2$,

$$\mathrm{F}(M) = 2\left(\frac{n}{n+1}\right)^{n+1},$$

so we need $\left(\frac{n}{n+1}\right)^{n+1} > \frac{1}{4}$, that is,

$$\left(1 + \frac{1}{n}\right)^{n+1} < 4.$$

? **Ask yourself** How does the given fact about $\left(1+\frac{1}{x}\right)^{x+1}$ help, when n could be anything from 2 upwards?

Because the function is decreasing for positive x, its value at any integer $n \geq 2$ is at most its value at $x = 2$:

$$\left(1 + \frac{1}{n}\right)^{n+1} \leq \left(\frac{3}{2}\right)^3 = \frac{27}{8} < 4.$$

Hence $\mathrm{F}(M) > \frac{1}{2}$, and the mode of X is greater than its median.

Putting the parts together, mean $<$ median $<$ mode for every $n > 1$.

Checks

- Take $n = 2$: the mean is $\frac{3}{5} = 0.6$ and the mode is $\frac{2}{3} \approx 0.667$. Solving $4x^3 - 3x^4 = \frac{1}{2}$ numerically gives median ≈ 0.6143, neatly in between. Also $\mathrm{F}(0.6) = 0.4752 < \frac{1}{2}$ and $\mathrm{F}(\frac{2}{3}) = \frac{16}{27} \approx 0.593 > \frac{1}{2}$.
- At $n = 2$ ($N = 3$) the expansion has exactly four terms, so the lower bound is an equality: both sides equal $\frac{125}{27} \approx 4.63$, comfortably above $6 - \frac{8}{5} = 4.4$.
- The ordering mean $<$ median $<$ mode was confirmed by computer for every n from 2 to 79.

Common slips

- Trying to find the median explicitly: the equation $x^{n+1}(n+2-(n+1)x) = \frac{1}{2}$ has no nice solution, and you don't need one.

- Getting the direction wrong: $\mu < m$ corresponds to $\mathrm{F}(\mu) < \frac{1}{2}$, because F is increasing.
- Forgetting to say why dropping the later binomial terms is allowed (they are all positive).

Beyond STEP

This density is a Beta distribution, Beta$(n+1, 2)$, one of the most important families on $[0, 1]$ and central to Bayesian statistics. The ordering mean $<$ median $<$ mode is typical of a distribution with a long tail to the left (negative skew), though it is not a universal law, which is why it had to be proved here. The decreasing function in part (iii) tends to e as $x \to \infty$, a classic limit that you will meet properly in analysis. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of the Beta distribution.

WORKED SOLUTION

STEP III 2022, Question 1

Pure • Roots of Polynomials

STEP III 2022, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Substituting the parametrisation into the circle's equation turns "where do the curves meet?" into "what are the roots of a quartic?", and from then on everything is done with the relationships between roots and coefficients, without ever solving the quartic. The useful extra trick is that the reciprocals $1/t_i$ are the roots of the same quartic with its coefficients written in reverse order. For the final part, "touching" means a repeated root.

Solution

Part (i)

The point with parameter t lies on C_2 exactly when

$$(ct - a)^2 + \left(\frac{c}{t} - b\right)^2 = r^2.$$

Expanding gives $c^2t^2 - 2act + a^2 + \dfrac{c^2}{t^2} - \dfrac{2bc}{t} + b^2 = r^2$. Since $t \neq 0$ we may multiply by t^2 without gaining or losing solutions:

$$c^2t^4 - 2act^3 + (a^2 + b^2 - r^2)t^2 - 2bct + c^2 = 0.$$

So each t_i is a root of $(*)$. Different values of t give different points (they have different x-coordinates ct), so the four points P_i give four distinct parameters. A quartic has at most four roots, so $t_1, \ldots, t_4$ are exactly the roots of $(*)$.

Part (ii)

Dividing $(*)$ by c^2, the sums of the roots taken one and two at a time are

$$\sum t_i = \frac{2a}{c}, \qquad \sum_{i<j} t_i t_j = \frac{a^2 + b^2 - r^2}{c^2}.$$

Using $\left(\sum t_i\right)^2 = \sum t_i^2 + 2\sum_{i<j} t_i t_j$,

$$\sum_{i=1}^{4} t_i^2 = \frac{4a^2}{c^2} - \frac{2(a^2+b^2-r^2)}{c^2} = \frac{2}{c^2}\left(a^2 - b^2 + r^2\right),$$

as required.

? **Ask yourself** How can we get at $\sum 1/t_i^2$ without a messy common denominator?

Put $u = 1/t$ in $(*)$. No root is zero (the constant term is $c^2 \neq 0$), so we can divide $(*)$ by t^4:

$$c^2 - 2acu + (a^2+b^2-r^2)u^2 - 2bcu^3 + c^2u^4 = 0,$$

that is, $c^2u^4 - 2bcu^3 + (a^2+b^2-r^2)u^2 - 2acu + c^2 = 0$. Its roots are the $1/t_i$. This is $(*)$ with a and b swapped, so the same calculation gives

$$\sum_{i=1}^{4} \frac{1}{t_i^2} = \frac{2}{c^2}\left(b^2 - a^2 + r^2\right).$$

Part (iii)

The point P_i is $(ct_i,\, c/t_i)$, so $OP_i^2 = c^2t_i^2 + \dfrac{c^2}{t_i^2}$. Summing and using part (ii),

$$\begin{aligned}\sum_{i=1}^{4} OP_i^2 &= c^2\sum t_i^2 + c^2\sum \frac{1}{t_i^2} \\ &= 2(a^2 - b^2 + r^2) + 2(b^2 - a^2 + r^2) \\ &= 4r^2,\end{aligned}$$

which is the required result. Notice that the centre (a, b) has completely disappeared.

Part (iv)

? **Ask yourself** What does "touch" mean in terms of the roots of $(*)$?

Let $f(t) = (ct-a)^2 + (c/t-b)^2 - r^2$, the squared distance from the moving point to the centre, minus r^2. The curve touches the circle at parameter t_0 when $f(t_0) = 0$ and $f'(t_0) = 0$: the point is on the circle and, to first order, its distance from the centre is not changing, so the curve's direction there is perpendicular to the radius, which is the circle's tangent direction. (This uses the fact that the velocity $(c, -c/t^2)$ is never zero.) The quartic in $(*)$ is $t^2 f(t)$, and since $t_0 \neq 0$, f has a double root at t_0 if and only if $(*)$ does.

So if C_1 and C_2 touch at two distinct points, with parameters $\alpha \neq \beta$, then α and β are both double roots of $(*)$. That accounts for all four roots, so

$$c^2t^4 - 2act^3 + (a^2+b^2-r^2)t^2 - 2bct + c^2 = c^2(t-\alpha)^2(t-\beta)^2.$$

Product of the roots. Comparing constant terms, $c^2\alpha^2\beta^2 = c^2$, so $(\alpha\beta)^2 = 1$ and $\alpha\beta = \pm 1$.

Sum of the roots. Comparing t^3 terms, $2(\alpha+\beta) = \dfrac{2a}{c}$, so $\alpha+\beta = \dfrac{a}{c}$.

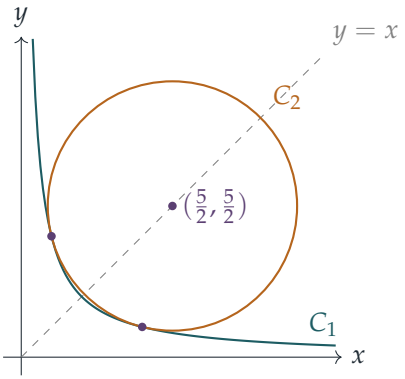
Sum of products of three. Comparing t terms, $2\alpha\beta(\alpha+\beta) = \dfrac{2b}{c}$, so $\alpha\beta(\alpha+\beta) = \dfrac{b}{c}$.

Dividing the last two (or substituting the second into the third) gives $b = \alpha\beta\, a$. Hence

$$\alpha\beta = 1 \implies b = a, \qquad \alpha\beta = -1 \implies b = -a,$$

so the centre (a, b) lies on $y = x$ or on $y = -x$. (If $a = 0$ then $b = 0$ too, and the centre is the origin, which lies on both lines.)

The figure shows the case $c = 1$, $\alpha = 2$, $\beta = \frac{1}{2}$. Matching coefficients gives $a = b = \frac{5}{2}$ and $r^2 = \frac{17}{4}$, and the circle touches the hyperbola at $(2, \frac{1}{2})$ and $(\frac{1}{2}, 2)$.



Checks

- In part (iii), take the circle $x^2 + y^2 = r^2$ centred at the origin with $c = 1$ and $r^2 > 2$. By symmetry the four points are $(p, q), (q, p), (-p, -q), (-q, -p)$, each at distance r, so the total really is $4r^2$.
- In the example in the figure, the squared distance from $(\frac{5}{2}, \frac{5}{2})$ to $(2, \frac{1}{2})$ is $\frac{1}{4} + 4 = \frac{17}{4} = 4.25$, as it should be. The radius direction $(-\frac{1}{2}, -2)$ is perpendicular to the curve's direction $(1, -\frac{1}{4})$ there, confirming tangency.
- The formulas in part (ii) were confirmed by computer for a few thousand random circles, by solving the quartic numerically.

Common slips

- Getting the sign of $\sum t_i t_j$ wrong. For $t^4 + pt^3 + qt^2 + \ldots$ the sum of the roots is $-p$ but the sum of the pairwise products is $+q$.

- In part (iv), using only the product of the roots. That gives $\alpha\beta = \pm 1$, which on its own says nothing about (a, b). One more coefficient comparison is needed to bring in a and b.
- Not saying why "touching" gives a double root. The reason is the condition $f(t_0) = f'(t_0) = 0$, together with $t_0 \neq 0$.

Beyond STEP

The relations between the roots and the coefficients are the elementary symmetric functions. A classical theorem says that *every* symmetric polynomial in the roots, such as $\sum t_i^2$ or $\sum t_i^{-2}$, can be written in terms of the coefficients. The reversal trick $u = 1/t$ is a first glimpse of projective ideas, and the fact that a conic and a circle meet in at most four points is a special case of Bézout's theorem on intersecting plane curves. For more on polynomials and their roots, see Allenby, *Rings, Fields and Groups*; for practice in this style, Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2022, Question 2

Pure • Infinite Descent

STEP III 2022, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 1.5, *Infinite Descent*: https://beyondstep.keithmcnulty.org/ch01/infinite-descent.html

The key idea

This is Fermat's *method of infinite descent*. Suppose a solution in non-zero integers exists. A divisibility argument shows that every variable shares a common factor. Dividing it out gives a new solution of the same kind (or of a closely related kind) in strictly smaller non-zero integers. That can't go on for ever, because positive integers can't decrease for ever. In each part the work is to find the right prime to look at (2 or 3) and to check that the new solution really has the same form as the old one.

Solution

Throughout we use two facts. If a prime m divides x^k then m divides x. And if x is a non-zero multiple of $m \geq 2$, then x/m is a non-zero integer with $|x/m| < |x|$.

Part (i)

Suppose $a^3 + 2b^3 + 4c^3 = 0$ with a, b, c non-zero integers. Then

$$a^3 = -2\left(b^3 + 2c^3\right)$$

is even. The cube of an odd number is odd, so a is even. Write $a = 2p$. Then p is a non-zero integer with $|p| = \frac{1}{2}|a| < |a|$. Substituting,

$$8p^3 + 2b^3 + 4c^3 = 0 \quad \Longrightarrow \quad 4p^3 + b^3 + 2c^3 = 0.$$

This is the first claim.

Now repeat the argument on the new equation. It gives $b^3 = -2\left(2p^3 + c^3\right)$, so b is

even, say $b = 2q$, and then

$$4p^3 + 8q^3 + 2c^3 = 0 \quad \Longrightarrow \quad 2p^3 + 4q^3 + c^3 = 0.$$

Once more, $c^3 = -2\left(p^3 + 2q^3\right)$, so $c = 2r$, and

$$2p^3 + 4q^3 + 8r^3 = 0 \quad \Longrightarrow \quad p^3 + 2q^3 + 4r^3 = 0.$$

Here $p = a/2$, $q = b/2$ and $r = c/2$ are non-zero integers with $|p| < |a|$, $|q| < |b|$ and $|r| < |c|$, as required.

Ask yourself Why does this mean there is no solution at all, rather than just "another solution"?

Suppose solutions in non-zero integers exist. Among all of them, choose one for which $|a|$ is as small as possible. This is possible because every non-empty set of positive integers has a least element. The argument above then produces another solution (p, q, r) in non-zero integers with $|p| < |a|$, which contradicts the minimality of $|a|$. So no such integers a, b, c exist. (Equivalently, the process would give an infinite strictly decreasing sequence $|a| > |a/2| > |a/4| > \cdots$ of positive integers, which is impossible.)

Part (ii)

This time the prime to use is 3. Suppose $9a^3 + 10b^3 + 6c^3 = 0$ with a, b, c non-zero integers.

- Since $10b^3 = -3\left(3a^3 + 2c^3\right)$, 3 divides $10b^3$. As 3 does not divide 10, it divides b^3, and so it divides b. Write $b = 3q$. Then $9a^3 + 270q^3 + 6c^3 = 0$, that is,

$$3a^3 + 90q^3 + 2c^3 = 0.$$

- Now $2c^3 = -3\left(a^3 + 30q^3\right)$, so $3 \mid c$. Write $c = 3r$: $3a^3 + 90q^3 + 54r^3 = 0$, that is,

$$a^3 + 30q^3 + 18r^3 = 0.$$

- Finally $a^3 = -3\left(10q^3 + 6r^3\right)$, so $3 \mid a$. Write $a = 3p$: $27p^3 + 30q^3 + 18r^3 = 0$, that is,

$$9p^3 + 10q^3 + 6r^3 = 0.$$

So $(p, q, r) = (a/3, b/3, c/3)$ is a solution of the *same* equation in non-zero integers with $|p| < |a|$. As in part (i), choose a solution with $|a|$ minimal. This gives a contradiction, so there are no non-zero integer solutions.

Part (iii)

Any integer that is not a multiple of 3 can be written as $3n \pm 1$ for some integer n. Then

$$(3n \pm 1)^2 = 9n^2 \pm 6n + 1 = 3\left(3n^2 \pm 2n\right) + 1,$$

which is one more than a multiple of 3. A multiple of 3, on the other hand, has a square that is a multiple of 3 (indeed of 9).

So each square leaves remainder 0 or 1 on division by 3, with remainder 0 exactly when the number itself is a multiple of 3. For $x^2 + y^2$, the possible remainders are

$$0 + 0 = 0, \qquad 0 + 1 = 1, \qquad 1 + 1 = 2.$$

So $x^2 + y^2$ is a multiple of 3 only when both remainders are 0, that is, only when both x and y are multiples of 3.

Now suppose $a^2 + b^2 = 3c^2$ with a, b, c non-zero integers. The right-hand side is a multiple of 3, so by the result just proved $a = 3p$ and $b = 3q$. Then

$$9p^2 + 9q^2 = 3c^2 \quad \Longrightarrow \quad c^2 = 3\left(p^2 + q^2\right).$$

So c^2 is a multiple of 3. From the first result, c must then be a multiple of 3, since otherwise c^2 would be one more than a multiple of 3. Write $c = 3r$:

$$9r^2 = 3\left(p^2 + q^2\right) \quad \Longrightarrow \quad p^2 + q^2 = 3r^2.$$

This is a solution of the same equation in non-zero integers with $|p| < |a|$. Choosing a solution with $|a|$ minimal gives the same contradiction as before, so no non-zero solutions exist.

Part (iv)

Here we work modulo 4. An even number $2m$ has square $4m^2 \equiv 0$, and an odd number $2m + 1$ has square $4m^2 + 4m + 1 \equiv 1 \pmod 4$. So a sum of three squares leaves remainder equal to the number of odd terms among the three (0, 1, 2 or 3). It is a multiple of 4 only when all three numbers are even.

> **Ask yourself** If we halve everything in $a^2 + b^2 + c^2 = 4abc$, do we get the same equation back?

No. If $a = 2p$, $b = 2q$ and $c = 2r$, the equation becomes $4(p^2 + q^2 + r^2) = 32pqr$, that is, $p^2 + q^2 + r^2 = 8pqr$. The coefficient has doubled. So we prove something stronger, which *is* preserved by halving:

Claim. For no integer $k \geq 2$ are there non-zero integers with $a^2 + b^2 + c^2 = 2^k abc$.

Suppose there were. Among all such solutions (with any $k \geq 2$), choose one with $|a|$ as small as possible. Since $k \geq 2$, the right-hand side $2^k abc$ is a multiple of 4. So $a^2 + b^2 + c^2$ is a multiple of 4, and by the remark above a, b and c are all even: $a = 2p$, $b = 2q$, $c = 2r$. Then

$$4\left(p^2 + q^2 + r^2\right) = 2^k \cdot 8pqr \quad \Longrightarrow \quad p^2 + q^2 + r^2 = 2^{k+1}\, pqr.$$

This is a solution in non-zero integers with exponent $k + 1 \geq 2$ and $|p| < |a|$, which contradicts the choice of a. So the claim holds. The case $k = 2$ shows that there are no non-zero integers a, b, c with $a^2 + b^2 + c^2 = 4abc$.

Checks

- A computer search over all non-zero a, b, c with $|a|, |b|, |c| \leq 40$ found no solution of any of the four equations, nor of $a^2 + b^2 + c^2 = 2^k abc$ for $2 \leq k \leq 6$.
- The power of 2 in the coefficient is what makes part (iv) work. Some equations of this shape *do* have solutions. For example, $1^2 + 1^2 + 1^2 = 3 \cdot 1 \cdot 1 \cdot 1$ and $3^2 + 3^2 + 3^2 = 1 \cdot 3 \cdot 3 \cdot 3$. There the coefficient is odd, so the mod-4 argument has nothing to bite on.
- Try the descent on a near miss: $a^2 + b^2 = 3c^2$ fails for $(a, b, c) = (1, 1, 1)$ because $1 + 1 = 2$ is not a multiple of 3, which is exactly the "$1 + 1 = 2$" case in part (iii).

Common slips

- Forgetting to say why the descent gives a contradiction. "We get smaller solutions" is not a proof until you invoke a minimal solution (or the fact that positive integers can't decrease for ever).
- In part (iv), halving and claiming the equation is unchanged. It isn't: the coefficient doubles, which is why the statement has to be strengthened.
- Writing "$3 \mid c^2$ so $3 \mid c$" without a reason. Here the reason is the first result of part (iii) (more generally, 3 is prime).

Beyond STEP

Infinite descent is how Fermat proved that $x^4 + y^4 = z^2$ has no solutions in positive integers, and so settled the $n = 4$ case of his Last Theorem. Each descent here amounts to showing that the equation has no solution modulo a suitable prime power in which the variables are not all divisible by that prime. This is the first step towards the p-adic numbers and the local–global ideas of modern number theory. The famous $a^2 + b^2 + c^2 = 3abc$ (the Markov equation) has infinitely many solutions, generated by a beautiful tree structure. Davenport's *The Higher Arithmetic* covers descent, congruences and sums of squares at exactly this level.

WORKED SOLUTION

STEP III 2022, Question 3

Pure • Implicit Differentiation

STEP III 2022, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Differentiate implicitly. A stationary point needs the numerator of $\frac{dy}{dx}$ to vanish, which gives y in terms of x (or a linear relation between them). Substituting back into the curve's equation leaves a single equation in x, and the question becomes: when does that equation have two real roots? In (i) it is a quadratic in disguise; in (ii) it is a quartic, and the way to count its roots is to find its turning points.

Solution

Part (i)

Differentiating $ax^2 + bxy + cy^2 = 1$ implicitly with respect to x,

$$2ax + by + (bx + 2cy)\frac{dy}{dx} = 0.$$

At a stationary point $\frac{dy}{dx} = 0$, so $2ax + by = 0$, that is $y = -\frac{2a}{b}x$ (we may divide by b because $abc \neq 0$). Substituting into the curve,

$$ax^2 - 2ax^2 + \frac{4a^2c}{b^2}x^2 = 1$$
$$\frac{a(4ac - b^2)}{b^2}x^2 = 1.$$

If $4ac - b^2 \leq 0$, the left-hand side is ≤ 0 for every x (recall $a > 0$), so there are no stationary points at all. So if the curve has two stationary points we must have $4ac - b^2 > 0$, that is $b^2 < 4ac$, as required. (Conversely, when $b^2 < 4ac$ we get $x = \pm|b|/\sqrt{a(4ac - b^2)}$, giving exactly two points.)

Ask yourself At these points, could $bx + 2cy$ also be zero, so that the derivative is not really defined?

If $2ax + by = 0$ and $bx + 2cy = 0$ together, then, since the determinant $4ac - b^2$ is non-zero, the only solution is $x = y = 0$, which is not on the curve. So the two points really do have $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.

Part (ii)

Differentiating $ay^3 + bx^2y + cx = 1$ implicitly,

$$(3ay^2 + bx^2)\frac{\mathrm{d}y}{\mathrm{d}x} + 2bxy + c = 0.$$

At a stationary point, therefore, $2bxy + c = 0$. As $c \neq 0$, $x \neq 0$, and so $y = -\dfrac{c}{2bx}$. Substituting into the curve,

$$-\frac{ac^3}{8b^3x^3} - \frac{cx}{2} + cx = 1, \qquad \text{i.e.} \qquad -\frac{ac^3}{8b^3x^3} + \frac{cx}{2} = 1.$$

Multiplying through by $8b^3x^3$ and rearranging,

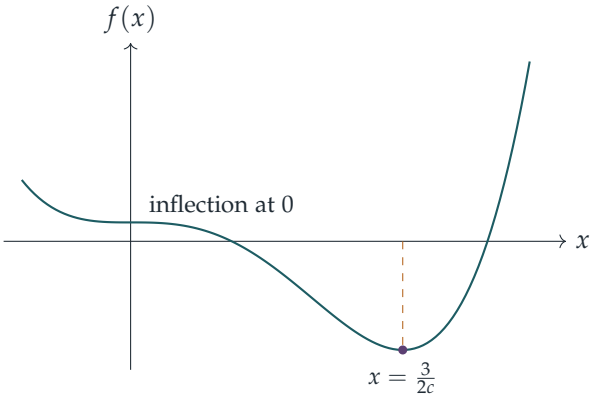
$$4cb^3x^4 - 8b^3x^3 - ac^3 = 0,$$

which is the required equation.

Call the left-hand side $f(x)$. Each stationary point has a different x-coordinate, because $y = -c/(2bx)$ is fixed by x. So two stationary points mean that f has at least two distinct real roots. To count the roots of f, look at its derivative:

$$f'(x) = 16cb^3x^3 - 24b^3x^2 = 8b^3x^2(2cx - 3).$$

The factor x^2 never changes sign, so f' changes sign only at $x = \frac{3}{2c}$. Suppose $c > 0$. Then the leading coefficient $4cb^3$ is positive (as $b > 0$), f is strictly decreasing for $x < \frac{3}{2c}$ and strictly increasing for $x > \frac{3}{2c}$; the point $x = 0$ is only a stationary point of inflection. Hence f has at most one root on each side of $\frac{3}{2c}$, and it has two distinct roots if and only if its minimum value is negative. If $c < 0$, everything is reversed: f increases then decreases, and has two distinct roots if and only if its maximum value is positive. In both cases, the condition is that $c\,f\left(\frac{3}{2c}\right) < 0$.



The case $a = -1$, $b = c = 1$: $f(x) = 4x^4 - 8x^3 + 1$ has a single turning point, a minimum at $x = \frac{3}{2}$, and two real roots.

Now evaluate at the turning point:

$$\begin{aligned} f\left(\tfrac{3}{2c}\right) &= 4cb^3 \cdot \frac{81}{16c^4} - 8b^3 \cdot \frac{27}{8c^3} - ac^3 \\ &= \frac{81b^3}{4c^3} - \frac{108b^3}{4c^3} - ac^3 = -\frac{27b^3 + 4ac^6}{4c^3}. \end{aligned}$$

So

$$c\, f\left(\tfrac{3}{2c}\right) = -\frac{4ac^6 + 27b^3}{4c^2},$$

and this is negative exactly when $4ac^6 + 27b^3 > 0$. Hence, if the curve has two stationary points, $4ac^6 + 27b^3 > 0$.

Part (iii)

The second and third equations say that both $2bxy + c$ and the coefficient $3ay^2 + bx^2$ of $\frac{\mathrm{d}y}{\mathrm{d}x}$ vanish, so this is a point of the curve where the method of (ii) breaks down. Since $c \neq 0$, the second equation gives $x \neq 0$ and $y = -\dfrac{c}{2bx}$. The first two equations are exactly what we used in (ii), so x satisfies

$$4cb^3x^4 - 8b^3x^3 - ac^3 = 0. \qquad (*)$$

Substituting $y = -c/(2bx)$ into the third equation,

$$\frac{3ac^2}{4b^2x^2} + bx^2 = 0 \quad \Longrightarrow \quad 4b^3x^4 = -3ac^2.$$

Multiplying by c, the first term of $(*)$ is $-3ac^3$, so $(*)$ becomes

$$-3ac^3 - 8b^3x^3 - ac^3 = 0 \quad \Longrightarrow \quad x^3 = -\frac{ac^3}{2b^3}.$$

Dividing $x^4 = -\dfrac{3ac^2}{4b^3}$ by this (allowed, as $x \neq 0$) gives

$$x = \frac{-3ac^2/(4b^3)}{-ac^3/(2b^3)} = \frac{3}{2c}.$$

So the solution sits exactly at the turning point of f from (ii). Putting $x = \frac{3}{2c}$ back into $x^3 = -\frac{ac^3}{2b^3}$:

$$\frac{27}{8c^3} = -\frac{ac^3}{2b^3} \quad \Longrightarrow \quad 54b^3 = -8ac^6 \quad \Longrightarrow \quad 4ac^6 + 27b^3 = 0,$$

as required. (Equivalently, $f\left(\frac{3}{2c}\right) = 0$ by the formula in (ii).) This is the boundary case of (ii): the two roots of the quartic have merged into a repeated root at the turning point, and the curve has a singular point there instead of two stationary points.

Checks

- In (i), take $a = c = 1$, $b = 1$: the curve $x^2 + xy + y^2 = 1$ is an ellipse, and $b^2 = 1 < 4$; its stationary points are at $x = \pm\frac{1}{\sqrt{3}}$, $y = \mp\frac{2}{\sqrt{3}}$, which do lie on it.
- In (ii), with $a = -1$, $b = c = 1$ we get $4ac^6 + 27b^3 = 23 > 0$ and f has minimum value $f(\frac{3}{2}) = -5.75 = -\frac{23}{4}$, matching $-\frac{27b^3+4ac^6}{4c^3}$.
- In (iii), choose $a = -2$, $c = 1.3$ and b with $27b^3 = -4ac^6$; then $x = \frac{3}{2c}$, $y = -\frac{c}{2bx}$ satisfies all three equations to rounding error. A computer check over thousands of random (a, b, c) also confirmed that the number of stationary points matches the conditions in (i) and (ii).

Common slips

- Forgetting that $x = 0$ is a critical point of the quartic: it is a point of inflection, not a turning point, and saying so is what lets you conclude there are at most two roots.
- Dropping the sign of c: the turning point is a minimum only when $c > 0$. Writing the condition as $c\,f(\frac{3}{2c}) < 0$ handles both cases at once.
- In (i), not noting that the case $b^2 = 4ac$ gives $0 = 1$, so it also has no stationary points.

Beyond STEP

The quantity $4ac^6 + 27b^3$ is, up to a factor, the *discriminant* of the quartic: it vanishes exactly when the quartic has a repeated root, just as $b^2 - 4ac$ does for a quadratic (and $4p^3 + 27q^2$ for the cubic $t^3 + pt + q$). Part (iii) shows the geometric meaning: a repeated root corresponds to a *singular point* of the curve, where both partial derivatives of $ay^3 + bx^2y + cx$ vanish together. Singular points and discriminants are the starting point of algebraic geometry; for a gentle, visual introduction to the related ideas about polynomials and curves, try Needham, *Visual Complex Analysis*, and for more problems in this style, Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2022, Question 4

Pure • Infinite Products

STEP III 2022, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.5, *Viète's Product*: https://beyondstep.keithmcnulty.org/ch02/viete.html

The key idea

Everything rests on the double-angle formula $\sinh 2\theta = 2\sinh\theta\cosh\theta$, applied over and over: each use halves the argument of the sinh and leaves behind a factor of cosh. After n halvings the leftover $\sinh(x/2^n)$ is tiny, and it behaves just like $x/2^n$, so letting $n \to \infty$ turns a finite identity into an infinite product. The last two parts are this product evaluated at well-chosen values of x, together with its trigonometric twin.

Solution

Part (i)

Write $P_n(x) = \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right)$. We prove that $\sinh x = 2^n P_n(x)\sinh\left(\frac{x}{2^n}\right)$ for every positive integer n.

Base case $n = 1$. By the double-angle formula with $\theta = x/2$, $\sinh x = 2\cosh\left(\frac{x}{2}\right)\sinh\left(\frac{x}{2}\right)$, which is the claim for $n = 1$.

Inductive step. Suppose the result holds for some $n = k$. Using the double-angle formula with $\theta = x/2^{k+1}$,

$$\sinh\left(\frac{x}{2^k}\right) = 2\cosh\left(\frac{x}{2^{k+1}}\right)\sinh\left(\frac{x}{2^{k+1}}\right),$$

so

$$\begin{aligned}\sinh x &= 2^k P_k(x)\sinh\left(\frac{x}{2^k}\right) = 2^k P_k(x)\cdot 2\cosh\left(\frac{x}{2^{k+1}}\right)\sinh\left(\frac{x}{2^{k+1}}\right)\\ &= 2^{k+1}P_{k+1}(x)\sinh\left(\frac{x}{2^{k+1}}\right),\end{aligned}$$

which is the result for $n = k+1$. By induction it holds for all positive integers n.

Deduction. If $x \neq 0$ then $x/2^n \neq 0$, so $\sinh(x/2^n) \neq 0$ (the only zero of sinh is at 0). We may therefore divide the identity by $x\sinh(x/2^n)/2^n$, which gives

$$\frac{\sinh x}{x} \cdot \frac{x/2^n}{\sinh(x/2^n)} = P_n(x),$$

as required.

Part (ii)

For $y \neq 0$, dividing the Maclaurin series by y gives

$$\frac{\sinh y}{y} = 1 + \frac{y^2}{3!} + \frac{y^4}{5!} + \cdots = 1 + y^2 \sum_{r=1}^{\infty} \frac{y^{2r-2}}{(2r+1)!}.$$

Ask yourself Is it enough to say "every term after the first tends to 0"?

Not quite: there are infinitely many terms, so we should bound their sum. For $0 < |y| \leq 1$ we have $y^{2r-2} \leq 1$, so

$$0 \leq \frac{\sinh y}{y} - 1 \leq y^2 \sum_{r=1}^{\infty} \frac{1}{(2r+1)!} = y^2(\sinh 1 - 1),$$

and the right-hand side tends to 0 as $y \to 0$. Hence $\dfrac{\sinh y}{y} \to 1$, and taking reciprocals, $\dfrac{y}{\sinh y} \to 1$.

Now fix $x \neq 0$ and let $n \to \infty$ in the identity from (i). Put $y = x/2^n$; then $y \to 0$, so the factor $\dfrac{x/2^n}{\sinh(x/2^n)}$ tends to 1. The left-hand side therefore tends to $\dfrac{\sinh x}{x}$, while the right-hand side is the partial product $P_n(x)$. So the partial products converge to $\dfrac{\sinh x}{x}$, which is exactly the statement

$$\frac{\sinh x}{x} = \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right)\cdots.$$

Part (iii)

With $x = \ln 2$ we have $e^{x/2^n} = 2^{1/2^n}$, so

$$\cosh\left(\frac{x}{2^n}\right) = \frac{2^{1/2^n} + 2^{-1/2^n}}{2} = \frac{2^{2/2^n} + 1}{2 \times 2^{1/2^n}} = \frac{1 + 2^{1/2^{n-1}}}{2 \times 2^{1/2^n}},$$

on multiplying top and bottom by $2^{1/2^n}$. For $n = 1$ this gives $\cosh\left(\frac{x}{2}\right) = \dfrac{1+2}{2\sqrt{2}} = \dfrac{3}{2\sqrt{2}}$, and for $n = 2$ it gives

$$\cosh\left(\frac{x}{4}\right) = \frac{1 + 2^{1/2}}{2 \times 2^{1/4}},$$

as required.

On the left of (ii), $\sinh(\ln 2) = \frac{1}{2}\left(2 - \frac{1}{2}\right) = \frac{3}{4}$, so $\dfrac{\sinh x}{x} = \dfrac{3}{4\ln 2}$.

On the right, the partial product of the first N factors is

$$P_N(\ln 2) = \frac{\prod_{n=1}^{N}\left(1 + 2^{1/2^{n-1}}\right)}{2^N \cdot 2^{\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^N}}} = \frac{3}{2} \cdot \prod_{k=1}^{N-1} \frac{1 + 2^{1/2^k}}{2} \cdot \frac{1}{2^{1-1/2^N}},$$

where we have taken out the $n = 1$ factor $(1+2)/2$, re-indexed with $k = n - 1$, and summed the geometric series in the exponent. As $N \to \infty$, $2^{1-1/2^N} \to 2$, and $P_N(\ln 2) \to \dfrac{3}{4\ln 2}$ by (ii). Hence the partial products $\prod_{k=1}^{N-1} \frac{1+2^{1/2^k}}{2}$ converge, and

$$\frac{3}{4\ln 2} = \frac{3}{2} \cdot \frac{1}{2} \cdot \prod_{k=1}^{\infty} \frac{1 + 2^{1/2^k}}{2},$$

and cancelling $\frac{3}{4}$,

$$\frac{1}{\ln 2} = \frac{1+2^{\frac{1}{2}}}{2} \times \frac{1+2^{\frac{1}{4}}}{2} \times \frac{1+2^{\frac{1}{8}}}{2} \times \cdots.$$

Part (iv)

The same argument works word for word with sin and cos, because $\sin 2\theta = 2\sin\theta\cos\theta$ has the same form. Induction gives

$$\sin x = 2^n \cos\left(\frac{x}{2}\right) \cdots \cos\left(\frac{x}{2^n}\right) \sin\left(\frac{x}{2^n}\right).$$

For $0 < x < \pi$ we have $0 < x/2^n < \pi$, so $\sin(x/2^n) \neq 0$ and we can divide as before. The Maclaurin series $\sin y = y - \frac{y^3}{3!} + \cdots$ gives, for $0 < |y| \leq 1$,

$$\left|\frac{\sin y}{y} - 1\right| \leq y^2 \sum_{r=1}^{\infty} \frac{1}{(2r+1)!} = y^2(\sinh 1 - 1) \to 0,$$

so $\sin y/y \to 1$ and, letting $n \to \infty$,

$$\frac{\sin x}{x} = \cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{4}\right)\cos\left(\frac{x}{8}\right)\cdots.$$

(Alternatively, put $x = \mathrm{i}t$ in (ii), using $\cosh(\mathrm{i}t) = \cos t$ and $\sinh(\mathrm{i}t) = \mathrm{i}\sin t$; but the series argument in (ii) was for real y, so redoing it directly is cleaner.)

Take $x = \frac{\pi}{2}$: the left-hand side is $\dfrac{1}{\pi/2} = \dfrac{2}{\pi}$, and the factors are $\cos\left(\frac{\pi}{4}\right)$, $\cos\left(\frac{\pi}{8}\right)$, $\cos\left(\frac{\pi}{16}\right)$, and so on.

Ask yourself How do we turn $\cos(\pi/8)$, $\cos(\pi/16)$, … into nested square roots?

Use $\cos 2\theta = 2\cos^2\theta - 1$ in the form $2\cos\theta = \sqrt{2+2\cos 2\theta}$, where we take the positive root because $\cos\theta > 0$ for $0 < \theta < \frac{\pi}{2}$. Let $c_n = 2\cos\left(\pi/2^{n+1}\right)$. Then $c_1 = 2\cos\frac{\pi}{4} = \sqrt{2}$, and with $\theta = \pi/2^{n+2}$,

$$c_{n+1} = \sqrt{2+c_n}.$$

So $c_1 = \sqrt{2}$, $c_2 = \sqrt{2+\sqrt{2}}$, $c_3 = \sqrt{2+\sqrt{2+\sqrt{2}}}$, and so on, and the nth factor of the product is $c_n/2$. Therefore

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2+\sqrt{2}}}{2} \times \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \times \cdots,$$

which is Viète's formula of 1593.

Checks

- In (i), put $n = 1$ and $x = 2\theta$: you recover the double-angle formula, as you should.
- In (iii), $\cosh(\frac{1}{2}\ln 2) = \frac{3}{2\sqrt{2}} \approx 1.0607$, and the formula for general n agrees with it at $n = 1$.
- Numerically, the product in (iii) settles quickly: the first 60 factors give $1.442695\ldots$, matching $1/\ln 2 = 1.4426950\ldots$; and the product in (iv) gives $0.6366197\ldots$, matching $2/\pi$. Both were confirmed by computer to about 15 significant figures.

Common slips

- Dividing by $\sinh(x/2^n)$ without noting that it is non-zero because $x \neq 0$.
- Off-by-one errors in (iii): the $n = 1$ factor is $3/(2\sqrt{2})$, not of the form $(1+2^{1/2^k})/2$, and the stray powers $2^{1/2}\cdot 2^{1/4}\cdots$ combine to give a factor of 2, not 1.
- Assuming "$y/\sinh y \to 1$" without any argument from the series, which is what the question asks for.

Beyond STEP

An infinite product $\prod a_n$ is defined as the limit of its partial products, just as an infinite sum is the limit of its partial sums; taking logarithms turns one into the other, so convergence of $\prod(1+u_n)$ is closely tied to convergence of $\sum u_n$. The product for $\sin x/x$ here sits alongside Euler's more famous product $\sin x = x\prod_{k\geq 1}\left(1 - x^2/(k^2\pi^2)\right)$, which leads to $\sum 1/k^2 = \pi^2/6$. For the careful treatment of limits that makes arguments like part (ii) rigorous, read Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2022, Question 5

Pure • Symmetric Definite Integrals

STEP III 2022, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

On a symmetric interval $[-a, a]$, the substitution $x \mapsto -x$ lets us fold the integral onto $[0, a]$, pairing the value at x with the value at $-x$. For integrands of the form $1/(1 + \text{something})$, the pair $\frac{1}{1+p} + \frac{1}{1+q}$ collapses to exactly 1 precisely when $pq = 1$, and $\mathrm{e}^x \cdot \mathrm{e}^{-x} = 1$ is the model case. The whole question is about spotting that pairing and then knowing when it is the *only* way the answer can come out to be a.

Solution

Part (i)

Let $I = \displaystyle\int_{-a}^{a} \frac{1}{1+\mathrm{e}^x}\,\mathrm{d}x$. Substituting $x = -u$ (so $\mathrm{d}x = -\mathrm{d}u$, and the limits swap back),

$$I = \int_{-a}^{a} \frac{1}{1+\mathrm{e}^{-u}}\,\mathrm{d}u = \int_{-a}^{a} \frac{\mathrm{e}^u}{\mathrm{e}^u+1}\,\mathrm{d}u,$$

after multiplying top and bottom by e^u. Adding this to the original form of I,

$$2I = \int_{-a}^{a} \frac{1+\mathrm{e}^x}{1+\mathrm{e}^x}\,\mathrm{d}x = \int_{-a}^{a} 1\,\mathrm{d}x = 2a,$$

so $I = a$ for every $a \geq 0$, as required.

Part (ii)

The explanation. Define $G(a) = \displaystyle\int_0^a g(x)\,\mathrm{d}x$ for $a \geq 0$. Since g is continuous, the fundamental theorem of calculus says G is differentiable with $G'(a) = g(a)$ (a one-sided derivative at $a = 0$). But G is identically zero on $[0, \infty)$, so its derivative is zero there. Hence $g(a) = 0$ for all $a \geq 0$. (Continuity matters: without it, g could be non-zero at isolated points without affecting any integral.)

Folding the integral. Split at 0 and put $x = -u$ in the left half:

$$\int_{-a}^{0} \frac{\mathrm{d}x}{1+\mathrm{f}(x)} = \int_{0}^{a} \frac{\mathrm{d}u}{1+\mathrm{f}(-u)}.$$

Writing $a = \int_0^a 1\,\mathrm{d}x$, we get

$$\int_{-a}^{a} \frac{\mathrm{d}x}{1+\mathrm{f}(x)} - a = \int_0^a g(x)\,\mathrm{d}x, \qquad \text{where} \quad g(x) = \frac{1}{1+\mathrm{f}(x)} + \frac{1}{1+\mathrm{f}(-x)} - 1.$$

Because $\mathrm{f} \geq 0$, both denominators are at least 1, so g is continuous.

Ask yourself The result just explained is a one-way statement. Which direction of the "if and only if" does it give, and where does the other come from?

- If $g(x) = 0$ for all $x \geq 0$, then $\int_0^a g = 0$ and the integral equals a for every $a \geq 0$. This direction is immediate.
- Conversely, if the integral equals a for every $a \geq 0$, then $\int_0^a g = 0$ for every $a \geq 0$, and the first paragraph (with this continuous g) gives $g(x) = 0$ for all $x \geq 0$.

So the integral condition holds if and only if $g(x) = 0$ for all $x \geq 0$.

The product condition. Write $p = \mathrm{f}(x)$, $q = \mathrm{f}(-x)$. Multiplying through by $(1+p)(1+q) > 0$,

$$\begin{aligned} \frac{1}{1+p} + \frac{1}{1+q} = 1 &\iff (1+q) + (1+p) = (1+p)(1+q) \\ &\iff 2 + p + q = 1 + p + q + pq \\ &\iff pq = 1. \end{aligned}$$

So $g(x) = 0$ for all $x \geq 0$ if and only if $\mathrm{f}(x)\mathrm{f}(-x) = 1$ for all $x \geq 0$. Finally, the expression $\mathrm{f}(x)\mathrm{f}(-x)$ is unchanged when x is replaced by $-x$, so it equals 1 for all $x \geq 0$ exactly when it equals 1 for all real x. That completes the chain of equivalences.

Part (iii)

Fold the integral exactly as before, now using $\mathrm{h}(-u) = \mathrm{h}(u)$ in the left half:

$$\begin{aligned} \int_{-a}^{a} \frac{\mathrm{h}(x)}{1+\mathrm{f}(x)}\,\mathrm{d}x &= \int_0^a \frac{\mathrm{h}(x)}{1+\mathrm{f}(x)}\,\mathrm{d}x + \int_0^a \frac{\mathrm{h}(-u)}{1+\mathrm{f}(-u)}\,\mathrm{d}u \\ &= \int_0^a \mathrm{h}(x)\left(\frac{1}{1+\mathrm{f}(x)} + \frac{1}{1+\mathrm{f}(-x)}\right)\mathrm{d}x \\ &= \int_0^a \mathrm{h}(x)\,\mathrm{d}x, \end{aligned}$$

because the bracket equals 1 whenever $\mathrm{f}(x)\mathrm{f}(-x) = 1$, by the equivalence shown in part (ii). (The conditions $\mathrm{f} \geq 0$ and continuity just make every integrand continuous.)

Part (iv)

? **Ask yourself** How can $\dfrac{e^{-x}}{\cosh x}$ be written in the form $\dfrac{\text{const}}{1+f(x)}$?

Since $\cosh x = \frac{1}{2}(e^x + e^{-x})$,

$$\frac{e^{-x}\cos x}{\cosh x} = \frac{2e^{-x}\cos x}{e^x + e^{-x}} = \frac{2\cos x}{1+e^{2x}},$$

multiplying top and bottom by e^x. Take $f(x) = e^{2x}$, which is continuous, positive, and satisfies $f(x)f(-x) = e^{2x}e^{-2x} = 1$; and take $h(x) = 2\cos x$, which is continuous and even. Part (iii) with $a = \frac{1}{2}\pi$ gives

$$\int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{e^{-x}\cos x}{\cosh x}\,dx = \int_0^{\frac{1}{2}\pi} 2\cos x\,dx = \Big[2\sin x\Big]_0^{\frac{1}{2}\pi} = 2.$$

Checks

- Part (i) is the special case $f(x) = e^x$, $h \equiv 1$ of part (iii), and gives $\int_0^a 1\,dx = a$, as it should.
- A non-example: $f(x) = x^2$ has $f(x)f(-x) = x^4 \neq 1$, and indeed $\int_{-1}^{1} \frac{dx}{1+x^2} = \frac{\pi}{2} \neq 1$.
- Evaluating the integral in part (iv) numerically with Simpson's rule gives 2.0000000; the identity in part (iii) was also confirmed by computer for many random choices of $f = e^{(\text{odd function})}$ and even h.

Common slips

- Proving only one direction of the "if and only if" in part (ii): the step from the integral condition back to $g = 0$ is exactly where the first result (and the continuity of g) is needed.
- Forgetting to say why $f(x)f(-x) = 1$ for $x \geq 0$ gives it for all x: the expression is symmetric under $x \mapsto -x$.
- In part (iv), trying to integrate $e^{-x}\cos x/\cosh x$ directly instead of rewriting it as $2\cos x/(1+e^{2x})$.

Beyond STEP

The argument in part (ii) is the fundamental theorem of calculus used "backwards", and it is the reason continuity hypotheses appear so often in analysis: integrals cannot see what happens at individual points, so only continuity lets you recover a function's values from its integrals. The folding trick is a cousin of splitting a function into its even and odd parts; here $\frac{1}{1+f(x)} - \frac{1}{2}$ is odd exactly when $f(x)f(-x) = 1$. Abbott's *Understanding Analysis* gives a careful treatment of the fundamental theorem and why continuity is needed.

WORKED SOLUTION

STEP III 2022, Question 6

Pure • Rolling Circles and Tangents

STEP III 2022, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Part (i) is a second-order Taylor expansion, and its real point is the warning in the last line: when the first-order terms vanish, the ratio is decided by the second-order terms. In part (ii) the rolling condition says that equal arcs of the two circles pass the point of contact, which tells us how far C_1 has turned. At the special point in (c) the first-order terms in the chord cancel exactly, so we need the second-order terms from (i).

Solution

Part (i)

For small α, $\cos\alpha \approx 1 - \frac{1}{2}\alpha^2$ and $\sin\alpha \approx \alpha$, with errors of order α^3. So

$$\begin{aligned}\cos(\theta+\alpha) - \cos\theta &= \cos\theta\cos\alpha - \sin\theta\sin\alpha - \cos\theta \\ &\approx \cos\theta\left(1 - \tfrac{1}{2}\alpha^2\right) - \alpha\sin\theta - \cos\theta = -\alpha\sin\theta - \tfrac{1}{2}\alpha^2\cos\theta,\end{aligned}$$

as required. In the same way,

$$\sin(\theta+\alpha) - \sin\theta \approx \alpha\cos\theta - \tfrac{1}{2}\alpha^2\sin\theta.$$

Dividing top and bottom of $(*)$ by α,

$$\frac{\sin(\theta+\alpha)-\sin\theta}{\cos(\theta+\alpha)-\cos\theta} \approx \frac{\cos\theta - \frac{1}{2}\alpha\sin\theta}{-\sin\theta - \frac{1}{2}\alpha\cos\theta} \longrightarrow -\frac{\cos\theta}{\sin\theta} = -\cot\theta,$$

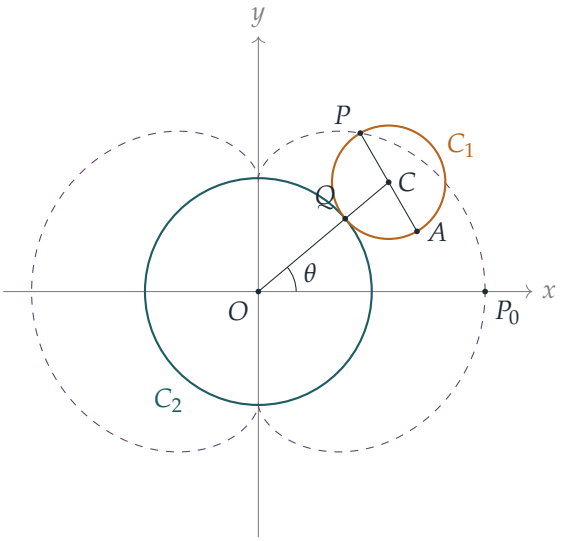
and the limit is fine because the denominator tends to $-\sin\theta \neq 0$.

If $\sin\theta = 0$ then $\cos\theta = \pm 1$, and the first-order term of the denominator disappears:

$$(*) \approx \frac{\alpha\cos\theta}{-\frac{1}{2}\alpha^2\cos\theta} = -\frac{2}{\alpha}.$$

So $(*)$ has no finite limit: its size grows without bound, tending to $-\infty$ as $\alpha \to 0^+$ and to $+\infty$ as $\alpha \to 0^-$. (Geometrically, the chord of the unit circle from angle θ to $\theta+\alpha$ becomes vertical, and its gradient blows up.)

Part (ii)(a)



The figure shows $n = 3$, with the dashed curve the path of P. Let C be the centre of C_1. Since the circles touch at Q, the points O, Q, C are collinear with $OC = (n-1)a + a = na$, so

$$\overrightarrow{OC} = na(\cos\theta, \sin\theta),$$

and from C the point Q is in direction $\theta + \pi$.

Let A be the point of C_1 diametrically opposite P; this is the point of C_1 that was touching C_2 at the start, at $((n-1)a, 0)$. Rolling without slipping means the arc of C_1 from A to Q equals the arc of C_2 that the contact has travelled, namely $(n-1)a\,\theta$. Since C_1 has radius a, the angle ACQ is $(n-1)\theta$.

Ask yourself Is A clockwise or anticlockwise from Q, as seen from C?

Rolling anticlockwise round the outside of C_2, the circle C_1 spins anticlockwise, so the points it has already rolled over are carried anticlockwise away from the contact point. So $\overrightarrow{CA}$ points in direction $\theta + \pi + (n-1)\theta = n\theta + \pi$, and $\overrightarrow{CP} = -\overrightarrow{CA}$ points in direction $n\theta$. (Check: at $\theta = 0$ this gives $P = ((n+1)a, 0)$. Altogether C_1 has turned through $n\theta$, the familiar "one extra turn" of a coin rolled round another.) Hence

$$\overrightarrow{OP} = na(\cos\theta, \sin\theta) + a(\cos n\theta, \sin n\theta),$$

so $x(\theta) = a(n\cos\theta + \cos n\theta)$, as required, and

$$y(\theta) = a(n\sin\theta + \sin n\theta).$$

Part (ii)(b)

Squaring and adding,

$$OP^2 = a^2\big(n^2 + 1 + 2n(\cos\theta\cos n\theta + \sin\theta\sin n\theta)\big) = a^2\big(n^2 + 1 + 2n\cos(n-1)\theta\big).$$

Setting this equal to $(n-1)^2a^2 = (n^2-2n+1)a^2$ gives $2n\cos(n-1)\theta = -2n$, that is $\cos(n-1)\theta = -1$. So

$$\theta = \frac{(2k+1)\pi}{n-1}, \qquad k \in \mathbb{Z},$$

giving $n-1$ distinct positions in each turn round C_2. These are the moments when P itself is the point of contact: the cusps of the curve.

Part (ii)(c)

With $\theta_0 = \pi/(n-1)$ we have $n\theta_0 = \theta_0 + \pi$, so

$$\cos n\theta_0 = -\cos\theta_0, \qquad \sin n\theta_0 = -\sin\theta_0.$$

Apply part (i) to each term, using $n\alpha$ as the small increment for the $n\theta$ terms:

$$\begin{aligned}
\frac{y(\theta_0+\alpha)-y(\theta_0)}{a} &\approx n\left(\alpha\cos\theta_0 - \tfrac{1}{2}\alpha^2\sin\theta_0\right) \\
&\quad + \left(n\alpha\cos n\theta_0 - \tfrac{1}{2}n^2\alpha^2\sin n\theta_0\right) \\
&= n\alpha\cos\theta_0 - \tfrac{1}{2}n\alpha^2\sin\theta_0 - n\alpha\cos\theta_0 + \tfrac{1}{2}n^2\alpha^2\sin\theta_0 \\
&= \tfrac{1}{2}n(n-1)\alpha^2\sin\theta_0, \\
\frac{x(\theta_0+\alpha)-x(\theta_0)}{a} &\approx n\left(-\alpha\sin\theta_0 - \tfrac{1}{2}\alpha^2\cos\theta_0\right) \\
&\quad + \left(-n\alpha\sin n\theta_0 - \tfrac{1}{2}n^2\alpha^2\cos n\theta_0\right) \\
&= \tfrac{1}{2}n(n-1)\alpha^2\cos\theta_0,
\end{aligned}$$

each with an error of order α^3. The first-order terms cancel exactly, just as in the $\sin\theta = 0$ case of (i), so the second-order terms decide the answer. Dividing top and bottom by $\frac{1}{2}n(n-1)\alpha^2$,

$$\frac{y(\theta_0+\alpha)-y(\theta_0)}{x(\theta_0+\alpha)-x(\theta_0)} = \frac{\sin\theta_0 + O(\alpha)}{\cos\theta_0 + O(\alpha)} \longrightarrow \tan\theta_0,$$

provided $\cos\theta_0 \neq 0$, which holds for $n \geq 4$. When $n = 3$, $\theta_0 = \frac{1}{2}\pi$; the same working shows the reciprocal ratio tends to $\cot\theta_0 = 0$, so the chord becomes vertical.

This limit is the gradient of the tangent at θ_0, because it is the limit of chord gradients. Now

$$x(\theta_0) = a(n\cos\theta_0 - \cos\theta_0) = (n-1)a\cos\theta_0, \qquad y(\theta_0) = (n-1)a\sin\theta_0,$$

so OP makes angle θ_0 with the x-axis and has gradient $\tan\theta_0$ (it is vertical when $n = 3$). The tangent and OP have the same direction, so the tangent at $(x(\theta_0), y(\theta_0))$ is parallel to OP. In fact it lies along the line OP, since it passes through P.

Checks

- For $\theta = 0$, $\alpha = 0.001$ the ratio $(*)$ is -1999.9998, against $-2/\alpha = -2000$.

- With $n = 5$, $\theta_0 = \frac{1}{4}\pi$ and $\alpha = 10^{-4}$, the chord gradient is 1.0004, close to $\tan\frac{1}{4}\pi = 1$.
- At θ_0, $OP = (n-1)a$, so the point of (c) is the $k = 0$ cusp found in (b). That is why the tangent there is radial.

Common slips

- Taking the rotation of C_1 to be $(n-1)\theta$ (the arc rolled) and forgetting the extra θ from C travelling round O.
- Using only first-order approximations in (c): they give $0/0$, which tells you nothing.
- Forgetting that $n = 3$ gives $\theta_0 = \frac{1}{2}\pi$, where $\tan\theta_0$ is undefined.

Beyond STEP

The curve is an epicycloid with $n-1$ cusps. At each cusp the velocity $(x'(\theta), y'(\theta))$ is zero, so the ordinary formula $\mathrm{d}y/\mathrm{d}x = y'/x'$ fails. Part (c) handles this by going to second order, which is l'Hôpital's rule in disguise. Points where a parametrised curve has zero velocity are called singular points, and they are studied in a first course on the differential geometry of curves. For the limits and Taylor approximations behind part (i), see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2022, Question 7

Pure • Vector Product Transformations

STEP III 2022, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 7.5, *Rotating in Three Dimensions*: https://beyondstep.keithmcnulty.org/ch07/rotations-3d.html

The key idea

Everything rests on splitting a vector into a part along $\mathbf{n}$ and a part perpendicular to $\mathbf{n}$. The map $\mathbf{r} \mapsto \mathbf{n} \times \mathbf{r}$ kills the parallel part and turns the perpendicular part through a right angle about $\mathbf{n}$ without changing its length. Once part (i) shows that applying it twice gives minus the perpendicular part, the maps in (ii) and (iii) can be read off by seeing what they do to each of the two parts separately.

Solution

Part (i)

With $\mathbf{n} = (a, b, c)^{\mathsf{T}}$ and $\mathbf{r} = (x, y, z)^{\mathsf{T}}$,

$$\mathrm{f}(\mathbf{r}) = \mathbf{n} \times \mathbf{r} = \begin{pmatrix} bz - cy \\ cx - az \\ ay - bx \end{pmatrix}.$$

Applying f again, the x-component of $\mathbf{n} \times \mathrm{f}(\mathbf{r})$ is b times the third component of $\mathrm{f}(\mathbf{r})$ minus c times the second:

$$\begin{aligned} b(ay - bx) - c(cx - az) &= aby - b^2x - c^2x + acz \\ &= -x(b^2 + c^2) + aby + acz, \end{aligned}$$

as required. Since $\mathbf{n}$ is a unit vector, $b^2 + c^2 = 1 - a^2$, so this becomes

$$-x + a^2x + aby + acz = a(ax + by + cz) - x = (\mathbf{n} \cdot \mathbf{r})\, a - x.$$

That is the x-component of $(\mathbf{n} \cdot \mathbf{r})\mathbf{n} - \mathbf{r}$. The y- and z-components follow in the same way by cycling $a \to b \to c \to a$ and $x \to y \to z \to x$, since the cross product is

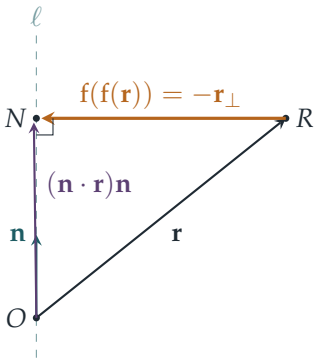
unchanged by this cyclic relabelling. Hence

$$\mathrm{f}(\mathrm{f}(\mathbf{r})) = (\mathbf{n}\cdot\mathbf{r})\mathbf{n} - \mathbf{r}.$$

Geometric meaning. Write $\mathbf{r} = \mathbf{r}_{\parallel} + \mathbf{r}_{\perp}$, where $\mathbf{r}_{\parallel} = (\mathbf{n}\cdot\mathbf{r})\mathbf{n}$ is the component along $\mathbf{n}$ and $\mathbf{r}_{\perp} = \mathbf{r} - (\mathbf{n}\cdot\mathbf{r})\mathbf{n}$ is perpendicular to $\mathbf{n}$. Then

$$\mathrm{f}(\mathrm{f}(\mathbf{r})) = -\mathbf{r}_{\perp}.$$

Let ℓ be the line through O in the direction of $\mathbf{n}$ and let N be the foot of the perpendicular from R to ℓ, so $\overrightarrow{ON} = \mathbf{r}_{\parallel}$. Then $\mathrm{f}(\mathrm{f}(\mathbf{r})) = \overrightarrow{ON} - \overrightarrow{OR} = \overrightarrow{RN}$: it is the vector from R straight to the axis ℓ, at right angles to it. Its length is the distance of R from ℓ.



Part (ii)

The axis. Since $\mathbf{n}\times\mathbf{n} = \mathbf{0}$, we have $\mathrm{f}(\mathbf{n}) = \mathbf{0}$ and so $\mathrm{f}(\mathrm{f}(\mathbf{n})) = \mathbf{0}$, giving $\mathrm{g}(\mathbf{n}) = \mathbf{n}$. By linearity $\mathrm{g}(\lambda\mathbf{n}) = \lambda\mathbf{n}$, so every point of the line ℓ is fixed.

Perpendicular vectors. Now take $\mathbf{r}$ with $\mathbf{n}\cdot\mathbf{r} = 0$. By part (i), $\mathrm{f}(\mathrm{f}(\mathbf{r})) = -\mathbf{r}$, so

$$\mathrm{g}(\mathbf{r}) = \mathbf{r} + \sin\theta\,(\mathbf{n}\times\mathbf{r}) - (1-\cos\theta)\mathbf{r} = \cos\theta\,\mathbf{r} + \sin\theta\,(\mathbf{n}\times\mathbf{r}).$$

? **Ask yourself** What does $\mathbf{n}\times\mathbf{r}$ look like when $\mathbf{r}\perp\mathbf{n}$?

It is perpendicular to both $\mathbf{n}$ and $\mathbf{r}$, so it lies in the plane through O perpendicular to $\mathbf{n}$, at right angles to $\mathbf{r}$. Its length is $|\mathbf{n}||\mathbf{r}|\sin 90^\circ = |\mathbf{r}|$. So $\mathbf{r}$ and $\mathbf{n}\times\mathbf{r}$ are two perpendicular vectors of equal length in that plane, and $\mathbf{n}\times\mathbf{r}$ is $\mathbf{r}$ turned through 90° in the right-handed sense about $\mathbf{n}$ (anticlockwise when viewed from the tip of $\mathbf{n}$ looking back towards O). Exactly as $\cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$ is $\mathbf{i}$ rotated by θ towards $\mathbf{j}$, the vector $\cos\theta\,\mathbf{r} + \sin\theta\,(\mathbf{n}\times\mathbf{r})$ is $\mathbf{r}$ rotated through θ about $\mathbf{n}$ in that same sense.

General vectors. f is linear (the cross product is linear in its second argument), so g is linear too. Writing a general $\mathbf{r}$ as $\mathbf{r}_{\parallel} + \mathbf{r}_{\perp}$, g leaves $\mathbf{r}_{\parallel}$ alone and rotates $\mathbf{r}_{\perp}$ through θ about $\mathbf{n}$, which is exactly what a rotation about ℓ does.

Therefore $R \mapsto P$ is a **rotation through angle θ about the line through O in the direction $\mathbf{n}$**, in the right-handed sense about $\mathbf{n}$ (anticlockwise when looking from the tip of $\mathbf{n}$ towards O).

Part (iii)

Using part (i),

$$\mathrm{h}(\mathbf{r}) = -\mathbf{r} - 2\big((\mathbf{n}\cdot\mathbf{r})\mathbf{n} - \mathbf{r}\big) = \mathbf{r} - 2(\mathbf{n}\cdot\mathbf{r})\mathbf{n}.$$

In terms of the two components, $\mathrm{h}(\mathbf{r}) = \mathbf{r}_\perp - \mathbf{r}_\parallel$:

- $\mathrm{h}(\mathbf{n}) = \mathbf{n} - 2\mathbf{n} = -\mathbf{n}$, so the component along $\mathbf{n}$ is reversed;
- if $\mathbf{n}\cdot\mathbf{r} = 0$ then $\mathrm{h}(\mathbf{r}) = \mathbf{r}$, so every point of the plane $\mathbf{n}\cdot\mathbf{r} = 0$ is fixed.

Geometrically, $\overrightarrow{RQ} = -2(\mathbf{n}\cdot\mathbf{r})\mathbf{n}$ is normal to the plane $\Pi:\ \mathbf{n}\cdot\mathbf{r} = 0$, and the midpoint of RQ has position vector $\mathbf{r} - (\mathbf{n}\cdot\mathbf{r})\mathbf{n}$, which lies in Π. So Q is the mirror image of R: the map $R \mapsto Q$ is **reflection in the plane through O perpendicular to $\mathbf{n}$**.

Checks

- Take $\mathbf{n} = \mathbf{k} = (0,0,1)^\mathsf{T}$. Then $\mathbf{n}\times\mathbf{r} = (-y,x,0)^\mathsf{T}$, twice gives $(-x,-y,0)^\mathsf{T}$, matching $(\mathbf{n}\cdot\mathbf{r})\mathbf{n} - \mathbf{r} = (0,0,z)^\mathsf{T} - (x,y,z)^\mathsf{T}$. Then $\mathrm{g}(\mathbf{r}) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta,\ z)^\mathsf{T}$, the familiar rotation about the z-axis, and $\mathrm{h}(\mathbf{r}) = (x,y,-z)^\mathsf{T}$, reflection in the plane $z = 0$.
- $\theta = 0$ gives $\mathrm{g}(\mathbf{r}) = \mathbf{r}$, and applying h twice gives the identity, as a rotation by nothing and a reflection should.
- A computer check with 2000 random unit vectors $\mathbf{n}$, vectors $\mathbf{r}$ and angles θ confirmed the x-component formula, the identity for $\mathrm{f}(\mathrm{f}(\mathbf{r}))$, that g agrees with a right-handed rotation by θ about $\mathbf{n}$, and that h agrees with reflection in the plane $\mathbf{n}\cdot\mathbf{r} = 0$.

Common slips

- Forgetting to use $a^2 + b^2 + c^2 = 1$: the identity in (i) is false for a non-unit $\mathbf{n}$.
- In (ii), only checking $\mathbf{r} \perp \mathbf{n}$ and not explaining why that, with $\mathrm{g}(\mathbf{n}) = \mathbf{n}$ and linearity, settles every $\mathbf{r}$.
- Calling (iii) a reflection in the *line* ℓ: points on ℓ are reversed, not fixed; it is the plane perpendicular to $\mathbf{n}$ that is fixed.

Beyond STEP

The formula in (ii) is Rodrigues' rotation formula, and the matrix of f is the antisymmetric "cross-product matrix" K, so a rotation is $I + \sin\theta\, K + (1-\cos\theta)K^2$, which is also $\exp(\theta K)$. The map in (iii) is a Householder reflection $I - 2\mathbf{n}\mathbf{n}^\mathsf{T}$, a workhorse of numerical linear algebra. In a first-year course these appear as the orthogonal matrices with determinant $+1$ and -1 respectively. Strang's *Introduction to Linear Algebra* is a good place to see them.

WORKED SOLUTION

STEP III 2022, Question 8

Pure • De Moivre and Irrationality

STEP III 2022, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 5.5, *Rational Angles That Aren't*: https://beyondstep.keithmcnulty.org/ch05/rational-angles.html

The key idea

De Moivre's theorem with $\cos\theta$ factored out gives $\sin k\theta$ and $\cos k\theta$ as $\cos^k\theta$ (or $\sin\theta\cos^{k-1}\theta$) times a polynomial in $\tan^2\theta = \sec^2\theta - 1$. When $\sec^2\theta - 1$ is an *even integer*, that polynomial is an integer whose parity we can read off from its first term. A rational angle in degrees would make some $\sin k\theta$ vanish, and the parity argument shows that this can never happen.

Solution

Part (i)

Assume $\cos\theta \neq 0$, so that $\sec\theta$ exists. By De Moivre's theorem,

$$\cos k\theta + \mathrm{i}\sin k\theta = (\cos\theta + \mathrm{i}\sin\theta)^k = \cos^k\theta\,(1+\mathrm{i}t)^k, \qquad t = \tan\theta.$$

Expanding binomially, $(1+\mathrm{i}t)^k = \sum_{j=0}^{k}\binom{k}{j}\mathrm{i}^j t^j$. The odd-$j$ terms are imaginary, with $\mathrm{i}^{2r+1} = (-1)^r\mathrm{i}$, and the even-$j$ terms are real, with $\mathrm{i}^{2r} = (-1)^r$. Taking imaginary parts,

$$\begin{aligned}\sin k\theta &= \cos^k\theta\left(\binom{k}{1}t - \binom{k}{3}t^3 + \binom{k}{5}t^5 - \cdots\right)\\ &= \cos^k\theta\,\tan\theta\left(k - \binom{k}{3}t^2 + \binom{k}{5}t^4 - \cdots\right).\end{aligned}$$

Since $\cos^k\theta\tan\theta = \sin\theta\cos^{k-1}\theta$ and $t^2 = \tan^2\theta = \sec^2\theta - 1$, this is

$$\sin k\theta = \sin\theta\cos^{k-1}\theta\left(k - \binom{k}{3}(\sec^2\theta - 1) + \binom{k}{5}(\sec^2\theta - 1)^2 - \cdots\right),$$

as required. The bracket stops of its own accord, because $\binom{k}{j} = 0$ for $j > k$. Taking real parts in the same way,

$$\cos k\theta = \cos^k\theta \left(1 - \binom{k}{2}(\sec^2\theta - 1) + \binom{k}{4}(\sec^2\theta - 1)^2 - \cdots\right).$$

Part (ii)

Here $\cos\theta = 1/a$ with $a \geq 3$, so $0^\circ < \theta < 90^\circ$. In particular $\sin\theta \neq 0$ and $\cos\theta \neq 0$. Also $\sec^2\theta - 1 = a^2 - 1$, which is an *even* integer because a is odd. Write $X = a^2 - 1$.

Suppose k is the least positive integer with $\sin k\theta = 0$. Then $k \neq 1$, since $\sin\theta \neq 0$, so $k > 1$ and part (i) applies:

$$0 = \sin k\theta = \sin\theta\cos^{k-1}\theta \left(k - \binom{k}{3}X + \binom{k}{5}X^2 - \cdots\right).$$

The factor $\sin\theta\cos^{k-1}\theta$ is non-zero, so the bracket is zero. Every term after the first contains at least one factor of the even integer X, and binomial coefficients are integers, so

$$0 = k - \binom{k}{3}X + \binom{k}{5}X^2 - \cdots \equiv k \pmod 2.$$

Hence k is even. Now

$$0 = \sin k\theta = 2\sin\left(\tfrac{1}{2}k\theta\right)\cos\left(\tfrac{1}{2}k\theta\right).$$

But $\frac{1}{2}k$ is an integer with $0 < \frac{1}{2}k < k$, so by the minimality of k, $\sin(\frac{1}{2}k\theta) \neq 0$. Therefore $\cos(\frac{1}{2}k\theta) = 0$, as required.

Ask yourself Where does the assumption that θ is rational come in?

It is what produces a k to start with. Suppose $\theta = p/q$ with p, q positive integers (recall θ is measured in degrees). Then $180q\,\theta = 180p$, so $\sin(180q\,\theta) = 0$. So there *is* a positive integer m with $\sin m\theta = 0$, and hence a least one; call it k. It satisfies the hypotheses above, so k is even and, writing $n = \frac{1}{2}k$, we have $\cos n\theta = 0$.

We now show that $\cos n\theta$ cannot be zero. If $n = 1$, then $\cos n\theta = 1/a \neq 0$. If $n > 1$, the cosine formula from part (i) gives

$$\cos n\theta = \frac{1}{a^n}\left(1 - \binom{n}{2}X + \binom{n}{4}X^2 - \cdots\right).$$

Again every term after the first is even, so the bracket is an odd integer, and in particular non-zero. So $\cos n\theta \neq 0$, a contradiction. Hence θ is irrational.

Part (iii)

Now $\cot\phi = 1/b$, so $\tan\phi = b$ with $b \geq 2$, and $0^\circ < \phi < 90^\circ$. Again $\sin\phi \neq 0$ and $\cos\phi \neq 0$, and this time

$$\sec^2\phi - 1 = \tan^2\phi = b^2,$$

which is an *even* integer because b is even. So exactly the argument of part (ii) runs with $X = b^2$ in place of $a^2 - 1$. In full:

Suppose $\phi = p/q$ with p, q positive integers. Then $\sin(180q\,\phi) = 0$, so there is a least positive integer k with $\sin k\phi = 0$, and $k > 1$ as $\sin\phi \neq 0$. By part (i), since $\sin\phi\cos^{k-1}\phi \neq 0$,

$$0 = k - \binom{k}{3}b^2 + \binom{k}{5}b^4 - \cdots \equiv k \pmod 2,$$

so k is even. Then $0 = \sin k\phi = 2\sin(\tfrac{1}{2}k\phi)\cos(\tfrac{1}{2}k\phi)$ and, by minimality, $\sin(\tfrac{1}{2}k\phi) \neq 0$, so $\cos n\phi = 0$ where $n = \tfrac{1}{2}k$. If $n = 1$ this says $\cos\phi = 0$, which is false. If $n > 1$,

$$\cos n\phi = \cos^n\phi\left(1 - \binom{n}{2}b^2 + \binom{n}{4}b^4 - \cdots\right),$$

where $\cos^n\phi \neq 0$ and the bracket is odd, so non-zero. This contradiction shows that ϕ is irrational.

Checks

- Try $k = 2$ and $k = 3$ in part (i): the formulas give $\sin 2\theta = 2\sin\theta\cos\theta$, $\cos 2\theta = \cos^2\theta - \sin^2\theta$ and $\sin 3\theta = \sin\theta(3\cos^2\theta - \sin^2\theta) = 3\sin\theta - 4\sin^3\theta$. Both formulas also agree with direct evaluation at random angles, for k up to 15.
- The parity conditions matter. With $a = 2$ (even) we get $\theta = 60°$, and with $b = 1$ (odd) we get $\phi = 45°$, both rational. In each case $X = 3$ or 1 is odd, so the argument breaks down exactly where it should.
- For $a = 3$, $\theta \approx 70.5288°$, and for $b = 2$, $\phi \approx 63.4349°$. A computer check confirms that $\sin k\theta$ and $\sin k\phi$ stay well away from zero for every $k < 2000$, for every odd a and even b below 30.

Common slips

- Forgetting to say why $\sin\theta\cos^{k-1}\theta \neq 0$ before dividing it out.
- Not explaining where k comes from: the minimal k exists only because a rational θ makes $\sin(180q\,\theta) = 0$.
- Concluding $\cos(\tfrac{1}{2}k\theta) = 0$ without using the minimality of k to rule out $\sin(\tfrac{1}{2}k\theta) = 0$.

Beyond STEP

The brackets here are, up to the factors in front, the *Chebyshev polynomials*, and the parity trick is a first taste of reducing integer identities modulo a prime. A famous relative is *Niven's theorem*: the only rational multiples θ of $180°$ for which $\cos\theta$ is rational are those with $\cos\theta \in \{0, \pm\tfrac{1}{2}, \pm 1\}$. It is proved by showing that $2\cos\theta$ is an algebraic integer.

Davenport's *The Higher Arithmetic* is a good place to build up the number theory behind arguments like this.

WORKED SOLUTION

STEP III 2022, Question 9

Mechanics • Collisions and Restitution

STEP III 2022, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Every collision here is the same collision: a moving particle hits a stationary one k times as heavy. So the factors α and β from the start of part (i) can be reused in part (ii), where B (mass km) hits C (mass k^2m). After that, each "meet at the same point" condition is just two expressions for the same time set equal to each other. In part (ii) we get two equations for k and e, and one of them turns out to be exactly the condition from part (i).

Solution

We model A, B, C as particles moving along one straight line on a smooth surface, with instantaneous collisions. All velocities are measured in the direction of A's initial motion.

Part (i)

The first collision. Momentum is conserved, and Newton's law of restitution says that the speed of separation is e times the speed of approach:

$$mu = mv_1 + kmv_2, \qquad v_2 - v_1 = eu.$$

Substituting $v_1 = v_2 - eu$ into the first equation gives $u = v_2 - eu + kv_2$, so

$$v_2 = \frac{1+e}{k+1}\,u = \beta u, \qquad v_1 = \frac{1+e}{k+1}\,u - eu = \frac{1+e-ek-e}{k+1}\,u = \frac{1-ke}{k+1}\,u = \alpha u,$$

as required.

The second collision. Measure time from the first collision. B covers the distance D to the wall in time $D/(\beta u)$. It rebounds with speed $e\beta u$, so it covers $\frac{1}{2}D$ back towards the collision point in a further time $D/(2e\beta u)$. If the second collision happens there, it happens at time

$$T = \frac{D}{\beta u}\left(1 + \frac{1}{2e}\right) = \frac{(2e+1)D}{2e\beta u}.$$

The point is $\frac{1}{2}D$ from the wall, so it is also $\frac{1}{2}D$ *beyond* the first collision point. A moves at the constant velocity αu, so it must have $\alpha > 0$ and $\alpha u T = \frac{1}{2}D$:

$$\alpha \frac{(2e+1)D}{2e\beta} = \frac{D}{2} \quad \Longleftrightarrow \quad \alpha(2e+1) = e\beta.$$

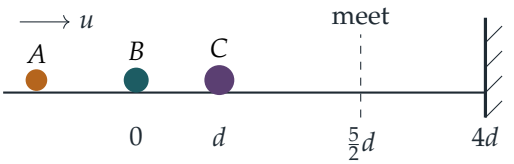
Multiplying by $k+1$ gives $(1-ke)(2e+1) = e(1+e)$, so

$$ke(2e+1) = (2e+1) - e - e^2 = 1 + e - e^2 \quad \Longrightarrow \quad k = \frac{1+e-e^2}{e(2e+1)},$$

which is the required result. This value does give $\alpha > 0$: it makes $ke = \frac{1+e-e^2}{2e+1}$, and this is less than 1 because $1 + e - e^2 < 1 + 2e$ for $e > 0$.

Part (ii)

Put the origin at B's starting point, so that C starts at $x = d$ and the wall is at $x = 4d$. The required meeting point is $\frac{3}{2}d$ from the wall, at $x = \frac{5}{2}d$.



Take $t = 0$ at the moment A hits B. That collision is exactly the one in part (i), so afterwards A moves with velocity αu and B with βu, where α and β are as before.

Ask yourself What happens when B hits C? Do we need to redo the collision algebra?

No. C is at rest and its mass $k^2 m$ is k times B's mass km. So this is the part (i) collision again, with mass ratio k, the same e, and incoming speed βu instead of u. B hits C at time $t_1 = d/(\beta u)$. Afterwards B moves at $\alpha\beta u$ and C at $\beta^2 u$.

Now we work out when each particle reaches $x = \frac{5}{2}d$, assuming no other collisions happen first (we check this at the end).

- A moves at αu from $x = 0$, so it gets there at $t = \dfrac{5d}{2\alpha u}$.
- B reaches $x = d$ at t_1 and then covers $\frac{3}{2}d$ at speed $\alpha\beta u$. It gets there at $t = \dfrac{d}{\beta u} + \dfrac{3d}{2\alpha\beta u}$.
- C covers $3d$ to the wall at $\beta^2 u$, then $\frac{3}{2}d$ back at $e\beta^2 u$. It gets there at

$$t = \frac{d}{\beta u} + \frac{3d}{\beta^2 u}\left(1 + \frac{1}{2e}\right) = \frac{d}{\beta u} + \frac{3(2e+1)d}{2e\beta^2 u}.$$

For a simultaneous collision all three times must be equal.

B and C. Cancel the common term $d/(\beta u)$ and multiply by $2\beta^2 u/(3d)$:

$$\frac{1}{\alpha}\beta = \frac{2e+1}{e} \quad \Longleftrightarrow \quad e\beta = \alpha(2e+1).$$

This is exactly the condition from part (i), so

$$k = \frac{1+e-e^2}{e(2e+1)}. \tag{1}$$

(This is no coincidence. In both situations, a particle leaves a collision, travels D to the wall at β times the incoming speed and comes back $\frac{1}{2}D$. Meanwhile the other particle travels $\frac{1}{2}D$ forward at α times the incoming speed. Here $D = 3d$, and the incoming speed is βu.)

A and B. Multiply $\dfrac{5}{2\alpha} = \dfrac{1}{\beta} + \dfrac{3}{2\alpha\beta}$ by $2\alpha\beta$ to get $5\beta = 2\alpha + 3$. Multiplying by $k+1$:

$$5(1+e) = 2(1-ke) + 3(k+1) \quad \Longrightarrow \quad 5e = k(3-2e) \quad \Longrightarrow \quad k = \frac{5e}{3-2e}. \tag{2}$$

Ask yourself We have two expressions for k. How do we get e?

We equate (1) and (2). Since $0 < e < 1$, both denominators are positive, so we can cross-multiply:

$$\begin{aligned} 5e^2(2e+1) &= (3-2e)(1+e-e^2) \\ 10e^3 + 5e^2 &= 3 + e - 5e^2 + 2e^3 \\ 0 &= 8e^3 + 10e^2 - e - 3. \end{aligned}$$

Trying $e = \frac{1}{2}$ gives $1 + \frac{5}{2} - \frac{1}{2} - 3 = 0$, so $2e - 1$ is a factor:

$$8e^3 + 10e^2 - e - 3 = (2e-1)(4e^2 + 7e + 3) = (2e-1)(4e+3)(e+1).$$

The other roots, $-\frac{3}{4}$ and -1, are negative, so the only root with $0 < e < 1$ is $e = \frac{1}{2}$, as required. Then $k = \frac{5}{4}$.

Checking the scenario is consistent. With $e = \frac{1}{2}$ and $k = \frac{5}{4}$ we get $\alpha = \frac{1}{6}$ and $\beta = \frac{2}{3}$. Before B hits C, A (speed $\frac{1}{6}u$) is slower than B (speed $\frac{2}{3}u$), so it cannot catch B early. After that, B (speed $\frac{1}{9}u$) is slower than C (speed $\frac{4}{9}u$), so B cannot catch C before C reaches the wall. B is now slower than A, so A gains on B, and they meet only at the point we computed. So there are no extra collisions, and all three particles reach $x = \frac{5}{2}d$ at time $t = 15d/u$.

Checks

- In part (i), $v_1 + kv_2 = u$ (momentum) and $v_2 - v_1 = eu$ (restitution). If $e = 1$ and $k = 1$, then $\alpha = 0$ and $\beta = 1$: equal masses swap velocities in a perfectly elastic collision, as they should.

- In part (ii), with $u = d = 1$ and the values above, the three arrival times are $\frac{5/2}{1/6} = 15$ for A, $\frac{1}{2/3} + \frac{3/2}{1/9} = \frac{3}{2} + \frac{27}{2} = 15$ for B, and $\frac{3}{2} + \frac{3}{4/9} + \frac{3/2}{2/9} = \frac{3}{2} + \frac{27}{4} + \frac{27}{4} = 15$ for C.
- Formula (1) at $e = \frac{1}{2}$ gives $k = \frac{5/4}{1} = \frac{5}{4}$, which agrees with (2): $\frac{5/2}{2} = \frac{5}{4}$.

Common slips

- Measuring the meeting point in part (i) from the wall when you need it from the collision point. Here they happen to be the same, $\frac{1}{2}D$, but in part (ii) the meeting point $\frac{3}{2}d$ from the wall is $\frac{5}{2}d$ from B's start.
- Redoing the B–C collision from scratch and losing a factor. Noticing that the mass ratio is again k saves the work.
- Forgetting that C comes back from the wall at $e\beta^2 u$ rather than $\beta^2 u$.

Beyond STEP

This question works because the problem is *self-similar*: scaling the masses by k and the speed by β turns one collision into the next, so the same algebra repeats. The same idea is behind the chains of balls of geometrically decreasing mass that pass energy along very efficiently, and behind the famous "counting π with colliding blocks" problem, where the analysis is cleanest in velocity space with momentum and energy as constraints. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions systematically, including working in the centre-of-mass frame, which makes formulas like α and β easy to see.

WORKED SOLUTION

STEP III 2022, Question 10

Mechanics • Elastic Springs Equilibrium

STEP III 2022, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Because the angle at P is a right angle, the two spring forces are perpendicular. Resolving along each spring therefore gives each tension on its own, with no simultaneous equations. For the energy, the elastic terms combine with the gravitational term so neatly that p becomes a simple function of θ. The real content of part (ii) is then the *physical* restriction on θ: both springs must be stretched.

Solution

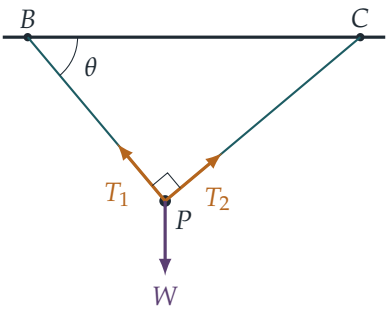
We model P as a particle and the springs as light and obeying Hooke's law, so a spring of modulus λ and natural length a stretched to length l has tension $\lambda(l-a)/a$ and stores energy $\lambda(l-a)^2/(2a)$.

Part (i)

Since $BC = 2a$ and angle $BPC = 90°$, triangle BPC has hypotenuse BC, so

$$PB = 2a\cos\theta, \qquad PC = 2a\sin\theta, \qquad \angle PCB = 90° - \theta.$$

Let T_1 and T_2 be the tensions in PB and PC.



The line PB makes angle θ with the horizontal, so it makes angle $90^\circ - \theta$ with the vertical; similarly PC makes angle θ with the vertical. As $PB \perp PC$, resolving along PB involves only T_1 and the weight, and the component of the weight along BP (away from B) is $W\cos(90^\circ - \theta) = W\sin\theta$. So

$$T_1 = W\sin\theta, \qquad \text{and likewise, along } PC, \qquad T_2 = W\cos\theta.$$

By Hooke's law,

$$T_1 = \frac{s_1 W(2a\cos\theta - a)}{a} = s_1 W(2\cos\theta - 1), \qquad T_2 = s_2 W(2\sin\theta - 1).$$

Equating,

$$s_1 = \frac{\sin\theta}{2\cos\theta - 1}, \qquad s_2 = \frac{\cos\theta}{2\sin\theta - 1},$$

which is the required s_1, together with s_2.

Part (ii)

Gravitational energy. P is below BC by the altitude of the triangle from P, which is $PB\sin\theta = 2a\sin\theta\cos\theta = a\sin 2\theta$. With zero level at BC, the gravitational potential energy is $-Wa\sin 2\theta$.

Elastic energy. The extension of PB is $a(2\cos\theta - 1)$, so, using part (i),

$$\frac{s_1 W\, a^2(2\cos\theta - 1)^2}{2a} = \tfrac{1}{2}Wa\, s_1(2\cos\theta - 1)^2 = \tfrac{1}{2}Wa\sin\theta\,(2\cos\theta - 1),$$

and similarly the energy in PC is $\frac{1}{2}Wa\cos\theta\,(2\sin\theta - 1)$.

Total. Adding,

$$\begin{aligned} -paW &= -Wa\sin 2\theta + \tfrac{1}{2}Wa\big(2\sin\theta\cos\theta - \sin\theta + 2\sin\theta\cos\theta - \cos\theta\big) \\ &= -Wa\sin 2\theta + Wa\sin 2\theta - \tfrac{1}{2}Wa(\sin\theta + \cos\theta), \end{aligned}$$

so

$$p = \tfrac{1}{2}(\sin\theta + \cos\theta) = \tfrac{\sqrt{2}}{2}\sin\big(\theta + \tfrac{\pi}{4}\big).$$

Ask yourself Can θ be anything between 0 and $\frac{\pi}{2}$?

No. The moduli $s_1 W$ and $s_2 W$ are positive, and the tensions $W\sin\theta$ and $W\cos\theta$ are positive, so both springs must be *stretched*: $2\cos\theta - 1 > 0$ and $2\sin\theta - 1 > 0$. (If either were zero, that spring would exert no force but would still need to supply a non-zero tension.) Hence $\cos\theta > \frac{1}{2}$ and $\sin\theta > \frac{1}{2}$, that is,

$$\tfrac{\pi}{6} < \theta < \tfrac{\pi}{3}, \qquad \text{so} \qquad \tfrac{5\pi}{12} < \theta + \tfrac{\pi}{4} < \tfrac{7\pi}{12}.$$

On this interval, which is symmetric about $\frac{\pi}{2}$, $\sin(\theta + \frac{\pi}{4})$ is at most 1 (attained at $\theta = \frac{\pi}{4}$) and strictly greater than its value at the endpoints, $\sin\frac{5\pi}{12} = \sin\frac{7\pi}{12}$. Now

$$\sin\tfrac{5\pi}{12} = \sin\big(\tfrac{\pi}{4} + \tfrac{\pi}{6}\big) = \tfrac{\sqrt{2}}{2}\cdot\tfrac{\sqrt{3}}{2} + \tfrac{\sqrt{2}}{2}\cdot\tfrac{1}{2} = \frac{\sqrt{2}(\sqrt{3}+1)}{4},$$

so

$$\tfrac{1}{2}\sqrt{2} \;\geq\; p \;>\; \tfrac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}(\sqrt{3}+1)}{4} = \tfrac{1}{4}(1+\sqrt{3}),$$

as required.

One significant figure. We need $0.65 \leq p < 0.75$, and we justify the surd estimates by squaring. Since $1.6^2 = 2.56 < 3$, we have $\sqrt{3} > 1.6$, so $p > \frac{1}{4}(2.6) = 0.65$. Since $1.5^2 = 2.25 > 2$, we have $\sqrt{2} < 1.5$, so $p \leq \frac{1}{2}\sqrt{2} < 0.75$. Hence $p = 0.7$ to one significant figure.

Checks

- At $\theta = \frac{\pi}{4}$ the set-up is symmetric, and indeed $s_1 = s_2 = \frac{1}{\sqrt{2}-1} = \sqrt{2}+1$, with $p = \frac{1}{2}\sqrt{2} \approx 0.7071$.
- As $\theta \to \frac{\pi}{6}^+$, $s_1 \to \infty$: the spring PB is at its natural length and must be infinitely stiff to carry a tension. Then $p \to \frac{1}{4}(1+\sqrt{3}) \approx 0.6830$, the lower bound.
- We confirmed by computer, for many random θ in $(\frac{\pi}{6}, \frac{\pi}{3})$, that the position found is a stationary point of the total potential energy (so really is an equilibrium) and that the energy equals $-\frac{1}{2}Wa(\sin\theta + \cos\theta)$.

Common slips

- Giving the gravitational energy a positive sign: P is *below* the zero level.
- Maximising $\frac{1}{2}(\sin\theta + \cos\theta)$ over all of $(0, \frac{\pi}{2})$ and so finding the wrong lower bound $\frac{1}{2}$; the springs must be in tension.
- Quoting $\sqrt{3} \approx 1.732$ without showing that the error cannot change the rounding.

Beyond STEP

The cancellation in part (ii) is no accident: at an equilibrium the total potential energy is stationary, and for Hooke's-law springs the elastic energy is exactly half of (tension $\times$ extension), which ties it to the work done against gravity. Studying equilibria as stationary points of potential energy, and their stability through the second derivative, is a central theme of first-year mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops this energy viewpoint carefully.

WORKED SOLUTION

STEP III 2022, Question 11

Probability and Statistics • Binomial Mean Deviation

STEP III 2022, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

A fair coin makes the distribution of X symmetric about its mean, so the probabilities below the mean match those above it. That symmetry turns the absolute value into a one-sided sum. The identity in (i)(c) then turns $\sum r\binom{2n}{r}$ into a tail sum of a smaller binomial distribution, which symmetry evaluates again.

Solution

Throughout, $X \sim \mathrm{B}(N, \frac{1}{2})$, so $\mathrm{P}(X = r) = \binom{N}{r}2^{-N}$ and $\mu = \frac{1}{2}N$. Because $\binom{N}{r} = \binom{N}{N-r}$, we have $\mathrm{P}(X = r) = \mathrm{P}(X = N - r)$: the distribution is symmetric about $\frac{1}{2}N$.

Part (i)(a)

With $N = 2n$, the map $r \mapsto 2n - r$ sends $\{0, \ldots, n-1\}$ to $\{n+1, \ldots, 2n\}$, so by symmetry

$$\mathrm{P}(X \leq n-1) = \mathrm{P}(X \geq n+1).$$

The three events $X \leq n-1$, $X = n$, $X \geq n+1$ cover every outcome, so $2\,\mathrm{P}(X \leq n-1) + \mathrm{P}(X = n) = 1$, which gives

$$\mathrm{P}(X \leq n-1) = \tfrac{1}{2}\big(1 - \mathrm{P}(X = n)\big).$$

Part (i)(b)

Here $\mu = n$, so

$$\delta = \sum_{r=0}^{2n} |r - n| \binom{2n}{r} \frac{1}{2^{2n}}.$$

The term $r = n$ is zero. Substituting $r = 2n - s$ in the terms with $r \geq n+1$ gives $|r - n| = n - s$ and $\binom{2n}{r} = \binom{2n}{s}$ with $0 \leq s \leq n-1$. So the terms above the mean exactly

repeat those below it:

$$\delta = 2\sum_{r=0}^{n-1}(n-r)\binom{2n}{r}\frac{1}{2^{2n}} = \sum_{r=0}^{n-1}(n-r)\binom{2n}{r}\frac{1}{2^{2n-1}},$$

as required.

Part (i)(c)

For $1 \le r \le 2n$,

$$r\binom{2n}{r} = \frac{r\,(2n)!}{r!\,(2n-r)!} = 2n\cdot\frac{(2n-1)!}{(r-1)!\,(2n-r)!} = 2n\binom{2n-1}{r-1}.$$

Now split the sum in (b) as $2^{2n-1}\delta = A - B$, where

$$A = n\sum_{r=0}^{n-1}\binom{2n}{r}, \qquad B = \sum_{r=1}^{n-1} r\binom{2n}{r}$$

(the $r = 0$ term of B is zero, so we drop it).

The sum A. By (a), $\sum_{r=0}^{n-1}\binom{2n}{r} = 2^{2n}\,\mathrm{P}(X \le n-1) = 2^{2n-1} - \frac{1}{2}\binom{2n}{n}$, so

$$A = n\,2^{2n-1} - \tfrac{n}{2}\binom{2n}{n}.$$

Ask yourself How do we add up $\binom{2n-1}{s}$ for s from 0 to $n-2$?

The sum B. By the identity, with $s = r - 1$,

$$B = 2n\sum_{s=0}^{n-2}\binom{2n-1}{s}.$$

The row $\binom{2n-1}{s}$, $0 \le s \le 2n-1$, has an even number of entries and adds up to 2^{2n-1}. It is symmetric under $s \mapsto 2n-1-s$, which swaps $\{0,\dots,n-1\}$ with $\{n,\dots,2n-1\}$, so the first half adds up to 2^{2n-2}. Our sum stops one short of that half, so

$$\sum_{s=0}^{n-2}\binom{2n-1}{s} = 2^{2n-2} - \binom{2n-1}{n-1} = 2^{2n-2} - \tfrac{1}{2}\binom{2n}{n},$$

using $\binom{2n}{n} = \binom{2n-1}{n-1} + \binom{2n-1}{n} = 2\binom{2n-1}{n-1}$. Hence

$$B = n\,2^{2n-1} - n\binom{2n}{n}.$$

Putting it together.

$$2^{2n-1}\delta = A - B = \left(n - \frac{n}{2}\right)\binom{2n}{n} = \frac{n}{2}\binom{2n}{n}, \qquad \text{so} \qquad \delta = \frac{n}{2^{2n}}\binom{2n}{n}.$$

Part (ii)

Now $N = 2n+1$ and $\mu = n + \frac{1}{2}$, which is not a possible value of X. The symmetry $r \mapsto 2n+1-r$ swaps $\{0,\dots,n\}$ with $\{n+1,\dots,2n+1\}$ and preserves both $|r-\mu|$ and $\binom{2n+1}{r}$. So, as in (i)(b),

$$\delta = 2\sum_{r=0}^{n}\left(n+\tfrac{1}{2}-r\right)\binom{2n+1}{r}\frac{1}{2^{2n+1}} = \frac{1}{2^{2n+1}}\sum_{r=0}^{n}(2n+1-2r)\binom{2n+1}{r}.$$

- By the same symmetry, the first half of the row $\binom{2n+1}{r}$ adds up to half of 2^{2n+1}: $\sum_{r=0}^{n}\binom{2n+1}{r} = 2^{2n}$.
- By the identity $r\binom{2n+1}{r} = (2n+1)\binom{2n}{r-1}$ (proved exactly as in (i)(c)), and then (i)(a),

$$\sum_{r=1}^{n} r\binom{2n+1}{r} = (2n+1)\sum_{s=0}^{n-1}\binom{2n}{s} = (2n+1)\left(2^{2n-1}-\tfrac{1}{2}\binom{2n}{n}\right).$$

Therefore

$$\sum_{r=0}^{n}(2n+1-2r)\binom{2n+1}{r} = (2n+1)2^{2n} - (2n+1)\left(2^{2n}-\binom{2n}{n}\right) = (2n+1)\binom{2n}{n},$$

and

$$\boxed{\delta = \frac{2n+1}{2^{2n+1}}\binom{2n}{n}}$$

when $N = 2n+1$.

Checks

- $N = 3$ ($n = 1$): $|X - \frac{3}{2}|$ is $\frac{3}{2}$ with probability $\frac{1}{4}$ and $\frac{1}{2}$ with probability $\frac{3}{4}$, so $\delta = \frac{3}{8}+\frac{3}{8} = \frac{3}{4}$. The formula gives $\frac{3\cdot 2}{8} = \frac{3}{4}$. With $n = 0$ ($N = 1$) it gives $\frac{1}{2}$, which is also right.
- $N = 4$ ($n = 2$): (i)(c) gives $\frac{2\cdot 6}{16} = \frac{3}{4}$, the same as for $N = 3$. In fact δ for $N = 2n+1$ always equals δ for $N = 2n+2$: since $\binom{2n+2}{n+1} = \frac{2(2n+1)}{n+1}\binom{2n}{n}$, the even formula with $n+1$ in place of n reduces to the odd one.
- Both formulas agree exactly with the definition of δ for every $N \le 60$, checked by computer; for example $N = 10$ gives $\delta = \frac{315}{256} \approx 1.230$.

Common slips

- Forgetting the factor 2 from the symmetry, which is where 2^{2n-1} rather than 2^{2n} comes from in (i)(b).
- Getting the range of s wrong after the shift $s = r-1$: the sum stops at $n-2$, one term short of half the row, and that missing term $\binom{2n-1}{n-1}$ is what makes the answer non-zero.

- In (ii), treating μ as an integer: no term vanishes, and both halves of the row have $n+1$ terms.

Beyond STEP

Stirling's formula gives $\binom{2n}{n} \approx 4^n/\sqrt{\pi n}$, so $\delta \approx \sqrt{N/(2\pi)}$ for large N. Since $\sigma = \frac{1}{2}\sqrt{N}$, this is $\delta \approx \sigma\sqrt{2/\pi}$, exactly the mean absolute deviation of a normal distribution, as the central limit theorem predicts. Grimmett and Welsh, *Probability: An Introduction*, covers both the normal approximation to the binomial and the central limit theorem behind it.

WORKED SOLUTION

STEP III 2022, Question 12

Probability and Statistics • Random Chords in a Circle

STEP III 2022, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.3, *What Does "Random" Mean?*: https://beyondstep.keithmcnulty.org/ch12/bertrand.html

The key idea

Every chord in this question comes from one fact: a chord at perpendicular distance d from the centre of a circle of radius a has length $2\sqrt{a^2-d^2}$. So the work is to find d in terms of the random quantities, then average. In part (ii) the substitution $u = a^2 - r^2\sin^2 t$ is natural because the density of R supplies exactly the factor r that the substitution needs.

Solution

Part (i)

Let Θ be the angle AOB, so Θ is uniform on $[0, 2\pi]$ with density $\frac{1}{2\pi}$. Triangle AOB is isosceles with apex angle Θ. Dropping the perpendicular from O to AB bisects both the angle and the chord, so

$$AB = 2a\sin\tfrac{1}{2}\Theta.$$

This holds for all $\Theta \in [0, 2\pi]$, since $\sin\frac{1}{2}\Theta \geq 0$ there; for $\Theta > \pi$ we are measuring the reflex angle, but the chord is the same. Hence

$$\mathrm{E}(AB) = \frac{1}{2\pi}\int_0^{2\pi} 2a\sin\tfrac{1}{2}\theta\,\mathrm{d}\theta = \frac{a}{\pi}\Big[-2\cos\tfrac{1}{2}\theta\Big]_0^{2\pi} = \frac{a}{\pi}(2+2) = \frac{4a}{\pi}.$$

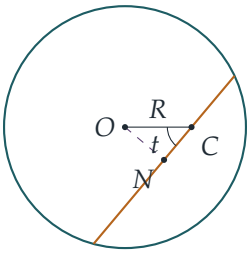
Part (ii)

Density of R. For $0 \le r \le a$, the event $R \le r$ means that C lies in the disc of radius r about O. Since probability is proportional to area,

$$\mathrm{P}(R \le r) = \frac{\pi r^2}{\pi a^2} = \frac{r^2}{a^2}, \qquad \text{so} \qquad f_R(r) = \frac{2r}{a^2} \quad (0 \le r \le a),$$

and $f_R(r) = 0$ otherwise.

Length of the chord. Let N be the foot of the perpendicular from O to the chord. In the right-angled triangle ONC, the hypotenuse is $OC = R$ and the angle at C is t, so $ON = R\sin t$.



The perpendicular from the centre bisects the chord, and N is its midpoint, so by Pythagoras in the triangle formed by O, N and an end of the chord,

$$\text{chord length} = 2\sqrt{a^2 - R^2\sin^2 t}.$$

Since $R < a$, the expression under the root is positive. Note that the whole chord is counted, on both sides of C.

The expected length. With t fixed and C random,

$$\mathrm{L}(t) = \int_0^a 2\sqrt{a^2 - r^2\sin^2 t}\,\frac{2r}{a^2}\,\mathrm{d}r.$$

Take $0 < t < \frac{1}{2}\pi$, so that $\sin t \neq 0$. Substitute $u = a^2 - r^2\sin^2 t$, so $\mathrm{d}u = -2r\sin^2 t\,\mathrm{d}r$; when $r = 0$, $u = a^2$, and when $r = a$, $u = a^2(1 - \sin^2 t) = a^2\cos^2 t$. Then

$$\mathrm{L}(t) = \frac{2}{a^2\sin^2 t}\int_{a^2\cos^2 t}^{a^2} u^{1/2}\,\mathrm{d}u = \frac{2}{a^2\sin^2 t}\cdot\frac{2}{3}\left(a^3 - a^3\cos^3 t\right) = \frac{4a(1-\cos^3 t)}{3\sin^2 t},$$

using $(a^2\cos^2 t)^{3/2} = a^3\cos^3 t$, which is valid because $\cos t \ge 0$ for an acute angle. This is the required formula.

Ask yourself How can the denominator $\sin^2 t$ be turned into something that cancels with $1 - \cos^3 t$?

Write $c = \cos t$. Then $\sin^2 t = 1 - c^2 = (1-c)(1+c)$, and $1 - c^3$ is a difference of cubes, $(1-c)(1+c+c^2)$. Since $c \neq 1$, we can cancel $1 - c$:

$$\mathrm{L}(t) = \frac{4a}{3}\cdot\frac{1+c+c^2}{1+c} = \frac{4a}{3}\left(c + \frac{1}{1+c}\right),$$

because $c(1+c)+1 = 1+c+c^2$. Finally, $1+\cos t = 2\cos^2 \frac{1}{2}t$, so $\dfrac{1}{1+\cos t} = \frac{1}{2}\sec^2 \frac{1}{2}t$, and

$$\mathrm{L}(t) = \frac{4a}{3}\Big(\cos t + \tfrac{1}{2}\sec^2\big(\tfrac{1}{2}t\big)\Big),$$

as required. This form has no trouble at $t = 0$: it gives $\frac{4a}{3}\cdot\frac{3}{2} = 2a$, which is right, since a chord through C along OC is a diameter.

Part (iii)

T has density $\frac{2}{\pi}$ on $[0, \frac{1}{2}\pi]$, so, using the second form of L (which is easy to integrate, since $\frac{\mathrm{d}}{\mathrm{d}t}\tan\frac{1}{2}t = \frac{1}{2}\sec^2\frac{1}{2}t$),

$$\begin{aligned}\mathrm{E}\big(\mathrm{L}(T)\big) &= \frac{2}{\pi}\cdot\frac{4a}{3}\int_0^{\pi/2}\Big(\cos t + \tfrac{1}{2}\sec^2\tfrac{1}{2}t\Big)\,\mathrm{d}t = \frac{8a}{3\pi}\Big[\sin t + \tan\tfrac{1}{2}t\Big]_0^{\pi/2}\\ &= \frac{8a}{3\pi}(1+1) = \frac{16a}{3\pi}.\end{aligned}$$

Checks

- At $t = \frac{1}{2}\pi$ the chord is perpendicular to OC, and $\mathrm{L} = \frac{4a}{3}(0+\frac{1}{2}\cdot 2) = \frac{4a}{3}$. Directly, $\int_0^a 2\sqrt{a^2-r^2}\,\frac{2r}{a^2}\,\mathrm{d}r = \frac{4a}{3}$ as well.
- $\mathrm{L}(t)$ should fall from $2a$ to $\frac{4}{3}a$ as t goes from 0 to $\frac{1}{2}\pi$, and the answer to (iii), $\frac{16}{3\pi}a \approx 1.698a$, lies between these, as an average must.
- With $a = 1.7$ and $t = 0.7$, numerical integration of the chord length against the density of R gives 3.01799, matching both formulae for $\mathrm{L}(0.7)$. The answer to (i), $\frac{4}{\pi}a \approx 1.273a$, was also confirmed by numerical integration.

Common slips

- Taking R to be uniform on $[0, a]$. Points are uniform by *area*, and there is more area far from the centre, hence the density $2r/a^2$.
- Using only the part of the chord on one side of C, or writing $ON = R\cos t$ (that is the distance CN along the chord).
- Integrating the first form of $\mathrm{L}(t)$ in part (iii) instead of the second form, which the question has set up to integrate in one line.

Beyond STEP

This is geometric probability, where "choose at random" only means something once a distribution is fixed. The answers $\frac{4a}{\pi}$ and $\frac{16a}{3\pi}$ differ because they describe different ways of picking a chord, the point behind Bertrand's paradox. Integral geometry makes this

systematic, using measures on lines that do not change under rotation and translation. For continuous distributions on regions and expectations of the kind computed here, see Blitzstein and Hwang, *Introduction to Probability*.

Beyond STEP

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